Numerical Methods. Lecture 5

The method of successive approximations and Newton's method on R: lecture, seminar and lab

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Introduction

Press the same key until the answer stops changing.

That is how many equations with no closed-form solution get solved: start from an approximation and improve it, step by step. The figure shows how the sequence closes in on the solution.

Here you learn why this works, when it works, how fast it works and how to program it.

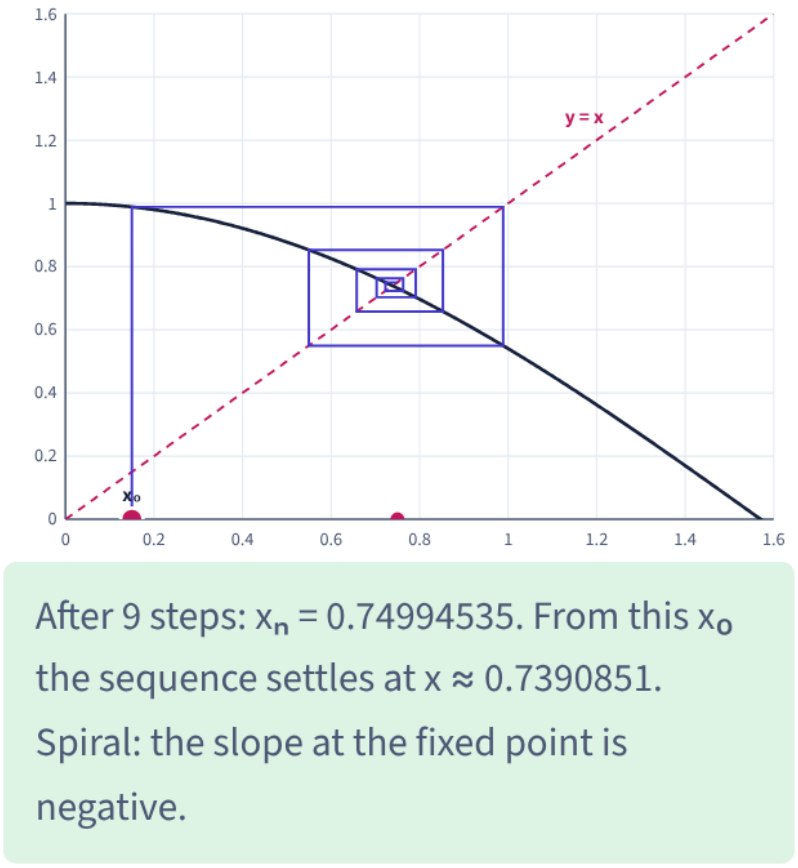


Diagram of the iteration x_{n+1} = cos x_n: the sequence closes in on the solution.

How to use this material

This is the Word version of the material: the interactive modules of the HTML version appear here as static figures, and solutions are shown directly. This material fully covers the learning unit "Lecture 5" (*4. The method of successive approximations on R* and *5. Newton's method on R. The tangent method*). Read it from top to bottom: each section builds on the one before. The four dots next to each title show its level.

BeginnerYou understand the ideas from pictures and examples, without heavy proofs.	**Intermediate**You apply the theorems and solve exam-style exercises on your own.	**Advanced**You prove the results and write robust programs.	**Top performance**You understand the limits of the methods, the order of convergence and acceleration.

If…	Go through
this is your first time with the subject	P1–P3, then A1–A5 and B1–B3, with all the figures in the lecture; come back to the proofs later
you are preparing for a seminar or exam	A5–A9, B3–B6, then the whole seminar S1–S7
you have a lab session	A5, A8, B1, B5, then L1–L8
you want top marks and more	all the proofs, the level 4 problems, L6–L9

Unit objectives

The source document has no explicit list of objectives. I derived them from its content, and at the end of this material (Z2) a matrix links each element of the source to the section that covers it.

By the end you will be able to:

1. define and recognize fixed points and α-contractions, geometrically and analytically;
2. prove the uniqueness of the fixed point, the derivative criterion and the existence of a fixed point;
3. turn $f(x)=0$ into $x=g(x)$ and check the hypotheses of the fixed-point theorem;
4. compute the number of iterations for a given accuracy and choose a stopping criterion;
5. derive Newton's formula and use its error estimates;
6. choose x_0 correctly with the tangent theorem and recognize failure situations;
7. implement, test and compare the methods in a program.

Before you start

Three short refreshers. If you already know them, go straight to A1.

P1. What solving an equation numerically means

Level: beginner

The equation $x^2 = 4$ has an exact answer. The equation $\cos x = x$ has no formula that gives its solution. Polynomial equations of degree 5 or higher cannot, in general, be solved by radicals either. Still, the solutions exist, and in practice we need them to a certain number of decimals.

An **iterative method** produces a sequence of approximations $x_0, x_1, x_2, \ldots$ that approaches the solution. We need three answers:

- **Convergence:** does the sequence surely approach the solution?
- **Speed:** how many steps are needed?
- **Stopping:** how do we know we can stop?

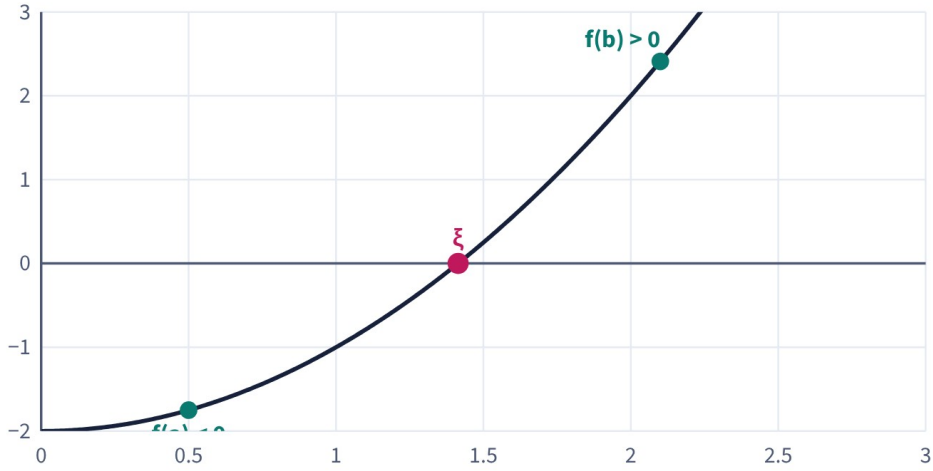
For the **absolute error** we use $|x_n - \xi|$, where ξ (read "xi") is the exact solution. Saying "with an error $\leq 10^{-4}$" means $|x_n - \xi| \leq 0.0001$. Since the solution is unknown, the error has to be *estimated*, not computed. This is where the theorems of this lecture come in.

P2. Three calculus theorems you need

Level: beginner

The intermediate value theorem

A continuous function cannot go from negative to positive values without passing through 0. If f is continuous on $[a, b]$ and $f(a) \cdot f(b) \leq 0$, there exists $\xi \in [a, b]$ with $f(\xi) = 0$.

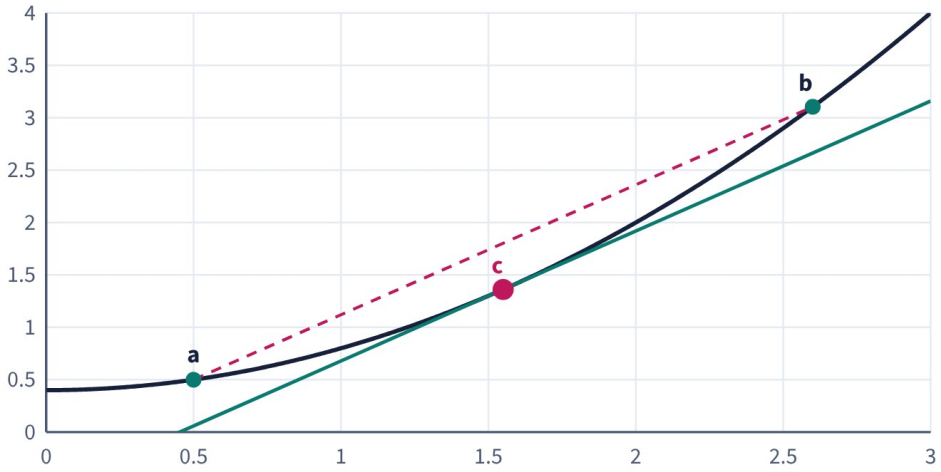


Example. For $f(x) = x^2 - 2$ we have $f(1) = -1 < 0$ and $f(2) = 2 > 0$, so there is a root in $[1, 2]$. It is $\sqrt{2}$.

Lagrange's theorem (the mean value theorem)

If g is continuous on $[x, y]$ and differentiable on (x, y), there is a point c between x and y where the tangent is parallel to the chord:

$$g(y)-g(x)=g'(c)(y-x).$$



Example. If we know that $|g'|\le 1/2$ everywhere, then $|g(y)-g(x)|\le 1/2|y-x|$: a small derivative means distances shrink. This is the central link of the lecture.

Taylor's formula of order 2

Near the point a, a function looks like its tangent, plus a correction:

$$f(x)=f(a)+f'(a)(x-a)+\frac{f''(c)}{2}(x-a)^2, c \text{ between } a \text{ and } x.$$

The first two terms are exactly the equation of the tangent. Newton starts from here.

P3. Rolle's sequence: how many roots are there?

Level: intermediate

The source uses the phrase "using Rolle's sequence" without details. The method says: *between two consecutive zeros of the derivative, the function has at most one root*, because it is monotonic on that interval.

1. Compute f' and find its zeros $c_1<c_2<\ldots<c_k$.
2. Build the table of values of f at $-\infty, c_1, \ldots, c_k, +\infty$ (limits at the ends).
3. Each sign change between two consecutive positions indicates exactly one root in that interval.

Example 1:* $f(x)=x^3+12x-1$*, from the lecture

$f'(x)=3x^2+12>0$ everywhere: the derivative has no zeros, so f is strictly increasing.

x	$-\infty$	$+\infty$
sign of f	−	+

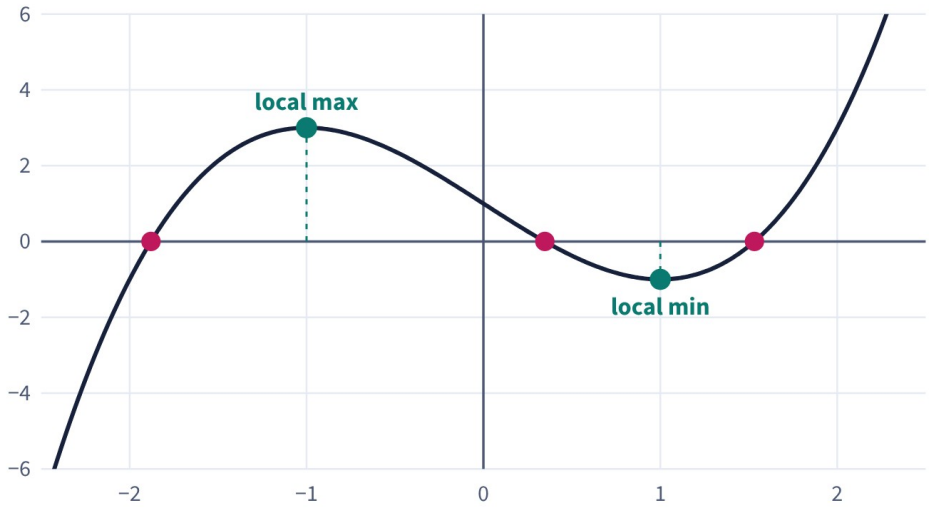
One sign change, so **one real root**. Since $f(0)=-1<0$ and $f(1)=12>0$, it lies in $[0,1]$.

Example 2: $f(x)=x^3-3x+1$

$f'(x)=3x^2-3=0$ gives $c_1=-1, c_2=1$. The values: $f(-1)=3, f(1)=-1$.

x	−∞	−1	1	+∞
sign of f	−	+	−	+

Three sign changes, so **three real roots**: one in each of the intervals $(-\infty,-1)$, $(-1,1)$ and $(1,+\infty)$.



Remember

Before any numerical method, **isolate the roots**: find how many there are and, for each one, an interval containing only that root. All the theorems of this lecture assume you have already done this.

Lecture: successive approximations

The simplest idea: rewrite the equation as $x = g(x)$ and repeat $x_{n+1} = g(x_n)$.

A1. The COS key experiment

Level: beginner

Take a scientific calculator set to radians. Enter any number and press the COS key, again and again:



The value after 40 presses of the COS key, starting from 1.

Whatever the start, you reach the same number: $0.7390851\ldots$ Without knowing it, the calculator has solved the equation $\cos x = x$.

Why? If the sequence $x_{n+1} = \cos x_n$ has the limit L, passing to the limit in this relation (cosine being continuous) gives $L = \cos L$. **The limit, if it exists, is a solution of the equation.** What remains is to know when the limit exists. That is the subject of this entire part.

> **A useful picture**
>
> Hold a map of the city above the city. Exactly one point on the map lies directly above the place it represents. The map "shrinks" distances, and a fixed point is guaranteed. This is, in essence, the theorem we are going to prove.

A2. Fixed point

Level: beginner

> **Definition 1 (fixed point)**
>
> Let $g : [a, b] \to \mathbb{R}$. A point $p \in [a, b]$ is called a **fixed point** of g if $g(p) = p$.

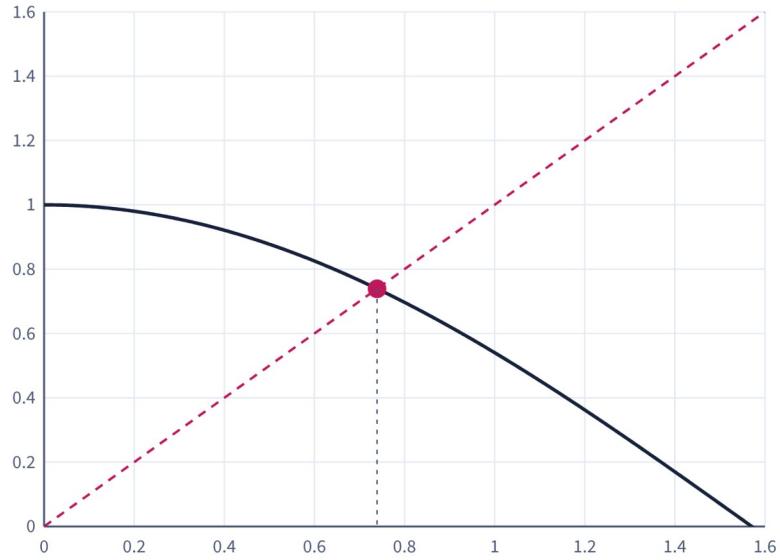
In other words, g does not move p. Three numerical examples:

- $g(x) = x^2$: $g(0) = 0$ and $g(1) = 1$, so 0 and 1 are fixed points. However, $g(2) = 4 \neq 2$.
- $g(x) = x/2 + 1$: solving $x/2 + 1 = x$ gives $p = 2$. Check: $g(2) = 2$.
- $g(x) = x + 1$: the equation $x + 1 = x$ has no solutions, so there are no fixed points.

The geometric picture (Remark 3 in the source)

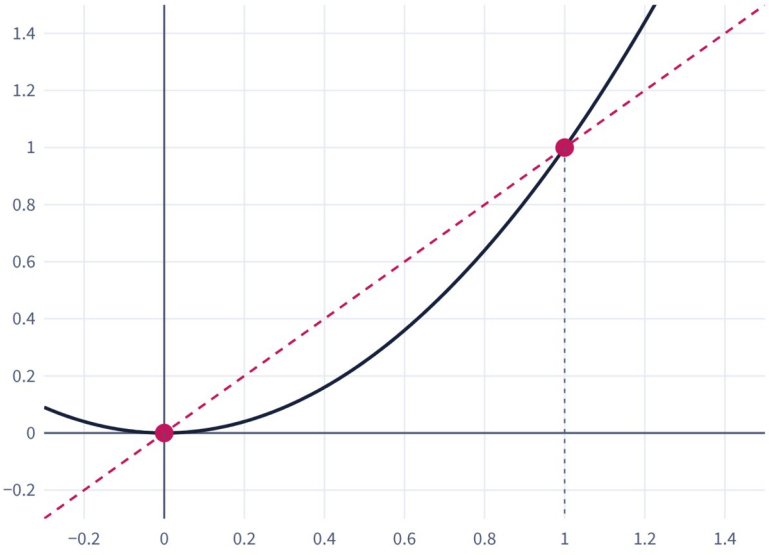
The condition $g(p) = p$ says that the point (p, p) lies on the graph of g and on the **first bisector** $y = x$. So: **a real root of the equation $x = g(x)$ is the abscissa of the intersection point of the curve $y = g(x)$ and the line $y = x$.**

How many fixed points does the function have?



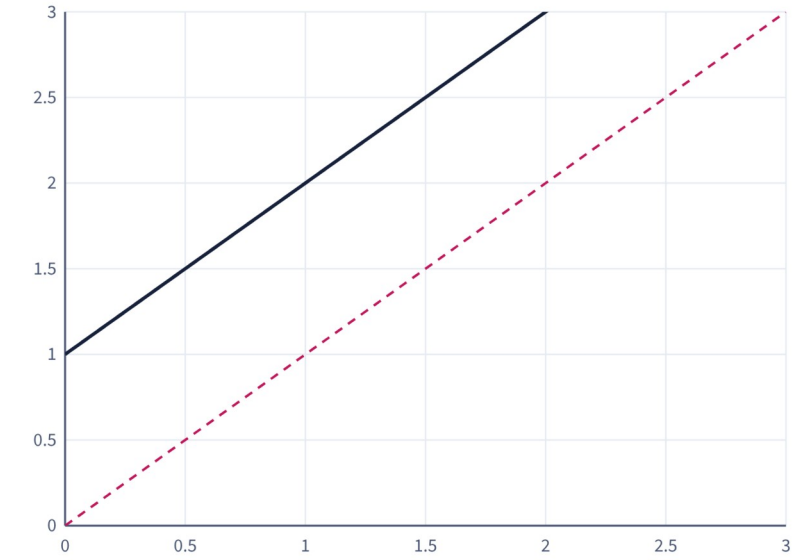
A single intersection, so a single fixed point.

g(x) = cos x: one fixed point.



Two fixed points: x = 0 and x = 1. Check: $0^2 = 0$ and $1^2 = 1$.

g(x) = x^2: two fixed points, 0 and 1.



The lines are parallel: there is no fixed point.

g(x) = x+1: no fixed point.

Which is a fixed point of $g(x)=x^2-2x$?

1. $x=2$
2. $x=3$
3. $x=1$

A3. Contractions: functions that bring points closer

Level: beginner

> **Definition 2 (α-contraction)**
>
> The function g is called an **α-contraction** on $[a,b]$ if there exists $\alpha \in (0,1)$ such that
>
> $$|g(x)-g(y)| \leq \alpha \cdot |x-y|, \ \forall\, x,y \in [a,b].$$

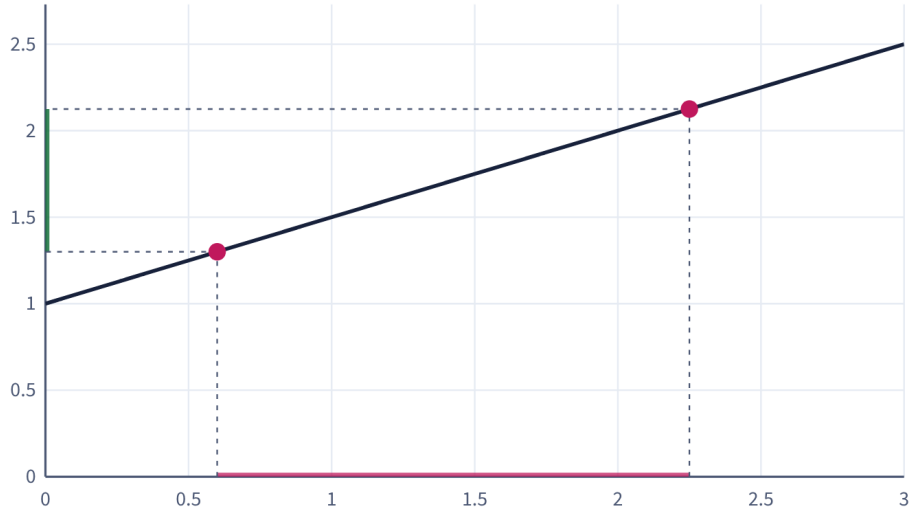
Read the inequality as follows: any two points, after applying g, end up closer, by at least the factor α. What matters is that the constant is **the same for all pairs**.

Complete example. $g(x)=x/2+1$. Take $x=0$, $y=2$: $g(0)=1$, $g(2)=2$. The distance was 2 and became 1, so the ratio is $1/2$. In general:

$$|g(x)-g(y)|=|x/2-y/2|=1/2|x-y|,$$

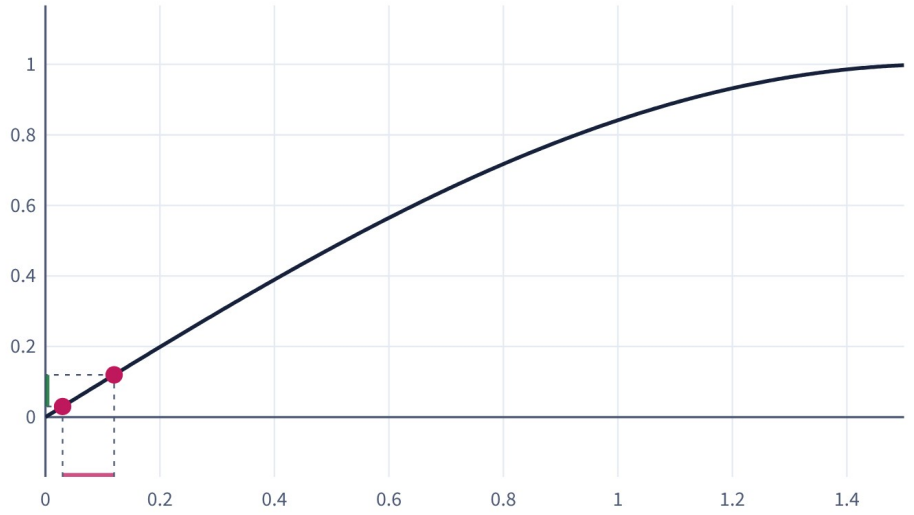
so g is a $1/2$-contraction.

Measure the shrinking ratio



Distance **|x−y| = 1.650** becomes **|g(x)−g(y)| = 0.825**. Ratio: **0.500**. The constant that would work for all pairs: 1/2.

g(x) = x/2+1: the ratio is always 1/2.



Distance **|x−y| = 0.090** becomes **|g(x)−g(y)| = 0.090**. Ratio: **0.997**. The constant that would work for all pairs: none exists, the ratio tends to 1 near 0.

g(x) = sin x: the ratio approaches 1 near 0.

Common mistake

It is not enough for the ratio to be less than 1 for *each* pair. For $\sin x$ on $[0, 1.5]$, the ratio is always below 1, but it gets arbitrarily close to 1 near 0. No $\alpha < 1$ works for all pairs. See the second figure above.

A4. Three fundamental remarks, with proofs

Level: intermediate

Remark 2: a small derivative guarantees contraction

Checking the definition on all pairs is hard. The derivative makes it easy, which is why we discuss it first.

> **Remark 2**
>
> If $g:[a,b]\to\mathbb{R}$ is differentiable and there exists $\alpha\in(0,1)$ such that $|g'(x)|\le\alpha$ for every $x\in[a,b]$, then g is an α-contraction.

1. Let $x,y\in[a,b]$, with $x<y$ (if $x=y$, the inequality is obvious). Being differentiable, g is continuous on $[x,y]$.
2. Apply Lagrange's theorem on $[x,y]$: there is $c\in(x,y)$ with $g(y)-g(x)=g'(c)(y-x)$.
3. Take absolute values: $|g(x)-g(y)|=|g'(c)|\cdot|x-y|$.
4. By hypothesis, $|g'(c)|\le\alpha$, so $|g(x)-g(y)|\le\alpha|x-y|$. This is exactly the definition. ■

Example. $g(x)=\cos x$ on $[0,1]$: $|g'(x)|=\sin x\le\sin 1\approx 0.8415<1$. So g is a contraction with $\alpha=\sin 1$.

Remark 1: at most one fixed point

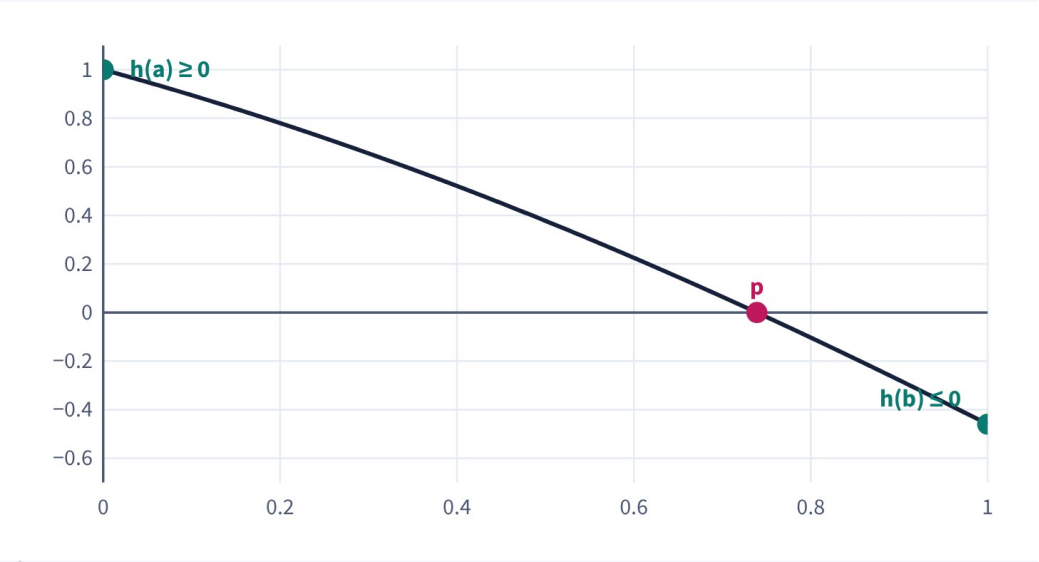
> **Remark 1**
>
> If g is an α-contraction and has a fixed point, then that point is unique.

1. Suppose there are two fixed points, p_1 and p_2: $g(p_1)=p_1$, $g(p_2)=p_2$.
2. From the definition of a contraction: $|g(p_1)-g(p_2)|\le\alpha|p_1-p_2|$. *Replace $g(p_i)$ with p_i on the left-hand side.*
3. We get $|p_1-p_2|\le\alpha|p_1-p_2|$, that is $|p_1-p_2|(1-\alpha)\le 0$.
4. Since $1-\alpha>0$, it follows that $|p_1-p_2|\le 0$, so $p_1=p_2$. ■ *Intuitively: the distance between two fixed points does not change, but a contraction would have to shrink it. The only possibility is distance zero.*

Remark 3: existence of a fixed point

> **Remark 3**
>
> If $g:[a,b]\to[a,b]$ is continuous, then g has a fixed point.

Note the **codomain** $[a,b]$: the graph stays inside the square $[a,b]\times[a,b]$. If it starts on the left side and reaches the right side, it must cross the diagonal. No contraction is required here.



The function h(x) = g(x) − x for g = cos, on [0, 1]: positive on the left, negative on the right.

1. Define $h:[a,b]\to\mathbb{R}$, $h(x)=g(x)-x$. The function h is continuous.
2. Since $g(a)\in[a,b]$, we have $g(a)\geq a$, so $h(a)=g(a)-a\geq 0$.
3. Since $g(b)\in[a,b]$, we have $g(b)\leq b$, so $h(b)=g(b)-b\leq 0$.
4. So $h(a)\cdot h(b)\leq 0$ and, h being continuous, there is $p\in[a,b]$ with $h(p)=0$, that is $g(p)=p$. ■

How they connect

Remark 3 gives **existence**, Remark 1 gives **uniqueness**, Remark 2 gives a **practical test** for contraction. Together, they lead to the theorem in A5.

A5. The theorem of successive approximations

Level: intermediate

Theorem (method of successive approximations)

Let $g:[a,b]\to[a,b]$ be an α-contraction, with $\alpha\in(0,1)$. Then, for every $x_0\in[a,b]$, the sequence $(x_n)_{n\in\mathbb{N}}$ given by

$$x_{n+1}=g(x_n),\ \forall\, n\in\mathbb{N},$$

converges to the unique fixed point p of g. Moreover, the following estimate holds:

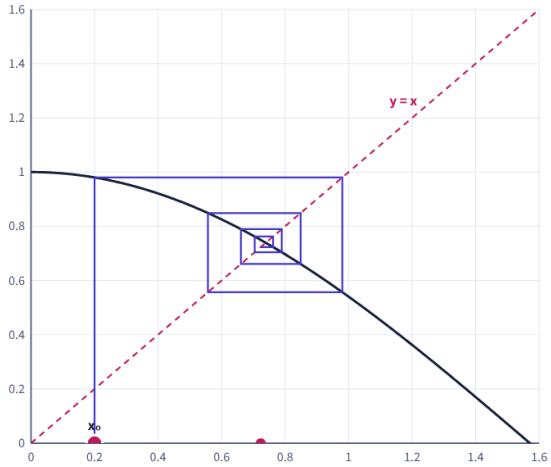
$$|x_n-p|\leq\alpha^n(b-a),\ \forall\, n\in\mathbb{N}^*.$$

The cobweb diagram

The iteration has a simple picture that you can construct with a ruler:

1. Start from x_0 on the Ox axis and go **vertically up to the curve** $y=g(x)$. You are now at height $g(x_0)=x_1$.
2. Go **horizontally to the bisector** $y=x$. The abscissa is now x_1.
3. Repeat.

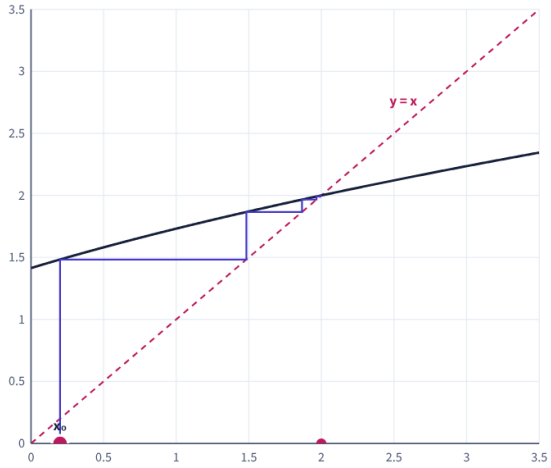
The iteration lab



n	x_n	$\lvert x_n - x_{n-1} \rvert$
0	0.20000000	—
1	0.98006658	0.78006658
2	0.55696725	0.42309933
3	0.84886217	0.29189491
4	0.66083755	0.18802461
5	0.78947844	0.12864089
6	0.70421571	0.08526272
7	0.76211956	0.05790385
8	0.72337417	0.03874539

After 8 steps: x_n = 0.72337417. From this x_0 the sequence settles at x ≈ 0.7390851. Spiral: the slope at the fixed point is negative.

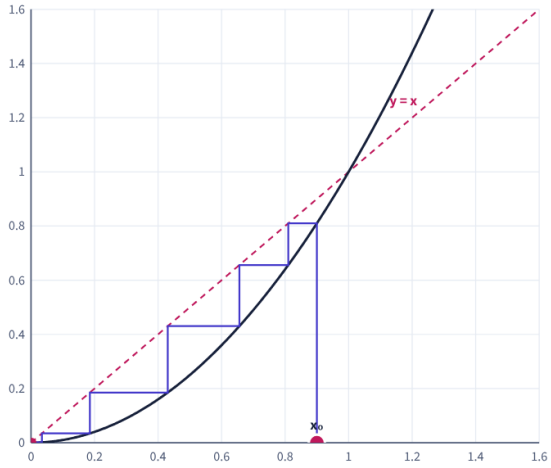
g(x) = cos x: spiral (negative slope at the fixed point).



n	x_n	$\lvert x_n - x_{n-1} \rvert$
0	0.20000000	—
1	1.48323970	1.28323970
2	1.86634394	0.38310424
3	1.96630210	0.09995816
4	1.99155771	0.02525561
5	1.99788831	0.00633061
6	1.99947201	0.00158370

After 6 steps: x_n = 1.99947201. From this x_0 the sequence settles at x ≈ 2.0000000. Staircase: the slope is positive.

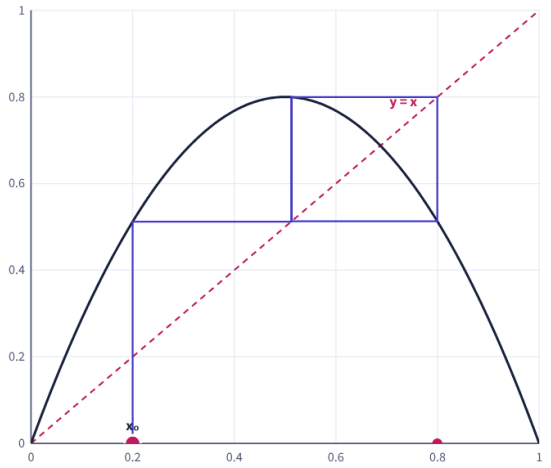
g(x) = √(x+2): staircase (positive slope).



n	x_n	$\lvert x_n - x_{n-1} \rvert$
0	0.90000000	–
1	0.81000000	0.09000000
2	0.65610000	0.15390000
3	0.43046721	0.22563279
4	0.18530202	0.24516519
5	0.03433684	0.15096518
6	0.00117902	0.03315782

After 6 steps: $x_n = 0.00117902$. From this x_0 the sequence settles at x ≈ 0.0000000. Not a contraction on [0, 1.6]. The point 1 repels, the point 0 attracts. Try $x_0 = 1.05$.

$g(x) = x^2$ from $x_0 = 0.9$: not a contraction on [0, 1.6]; 1 repels, 0 attracts.



n	x_n	$\lvert x_n - x_{n-1} \rvert$
0	0.20000000	–
1	0.51200000	0.31200000
2	0.79953920	0.28753920
3	0.51288406	0.28665514
4	0.79946880	0.28658475
5	0.51301899	0.28644981
6	0.79945762	0.28643862
7	0.51304043	0.28641719
8	0.79945583	0.28641540
9	0.51304386	0.28641197
10	0.79945554	0.28641169
11	0.51304441	0.28641114
12	0.79945550	0.28641109

After 12 steps: $x_n = 0.79945550$. From this x_0 the sequence does not settle: it oscillates or wanders. $|g'(p)| > 1$: the sequence does not tend to p, it jumps between two values.

$g(x) = 3.2 \cdot x(1-x)$: $|g'(p)| > 1$, the sequence ends up in a cycle.

Positive slope, $0 < g'(p) < 1$

The diagram looks like a **staircase**: the sequence approaches from one side, monotonically. See $\sqrt{x+2}$.

Negative slope, $-1 < g'(p) < 0$

The diagram looks like a **spiral**: the sequence jumps alternately to either side of p. See $\cos x$.

The proof

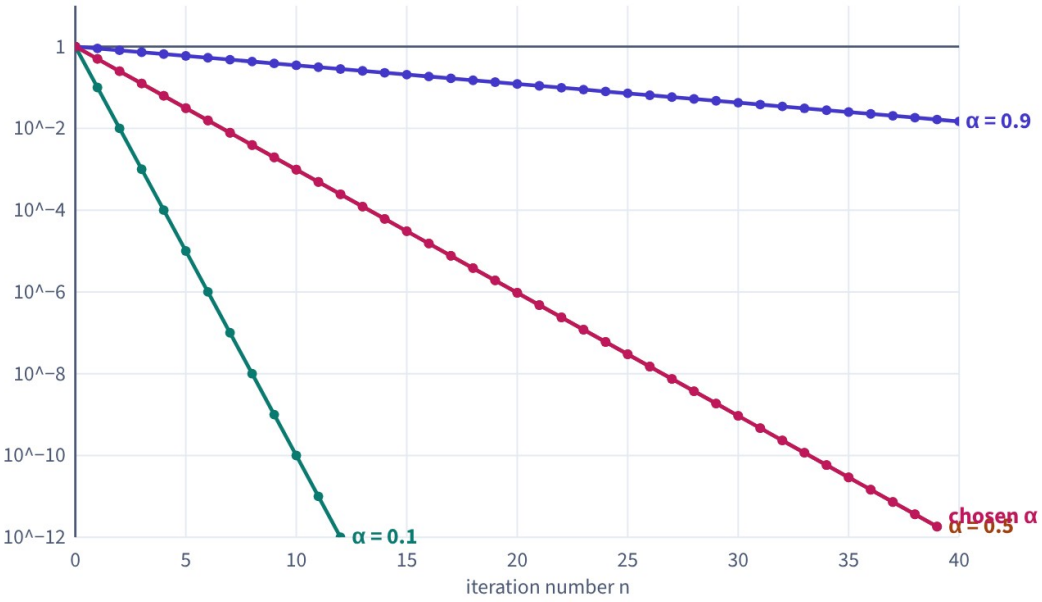
The source states the theorem without proof, but refers to it ("from the proof of Banach's theorem"). Here it is in full.

1. **The sequence is well defined.** Since g maps $[a,b]$ into $[a,b]$, all terms x_n stay in $[a,b]$ and g can always be applied.
2. **The fixed point exists and is unique.** A contraction is continuous: if $x_k \to x$, then $|g(x_k)-g(x)| \le \alpha |x_k - x| \to 0$. Remark 3 gives the existence of p, and Remark 1 gives uniqueness.
3. **One step brings us closer by the factor α.** $|x_n - p| = |g(x_{n-1}) - g(p)| \le \alpha |x_{n-1} - p|$. *We use $p = g(p)$ and the definition of a contraction.*
4. **Iterate the inequality.** $|x_n - p| \le \alpha |x_{n-1} - p| \le \alpha^2 |x_{n-2} - p| \le \ldots \le \alpha^n |x_0 - p|$.
5. **Bound the initial distance.** x_0 and p are in $[a,b]$, so $|x_0 - p| \le b - a$. Thus $|x_n - p| \le \alpha^n (b-a)$.
6. **Convergence.** Since $0 < \alpha < 1$, we have $\alpha^n \to 0$. So $|x_n - p| \to 0$, that is $x_n \to p$. ■

Speed depends on α (Remark 4)

The smaller α is, the faster the convergence. The error decreases like a geometric progression.

How much α matters



For α = 0.50, shrinking the distance a million times takes **20 iterations**. Each iteration gains 0.30 correct decimal digits.

A6. How close are we? Error estimates

Level: intermediate

The source gives two a priori bounds (computable *before* iterating). We add a third, a posteriori bound (computed *during* the iteration), which is very useful in practice.

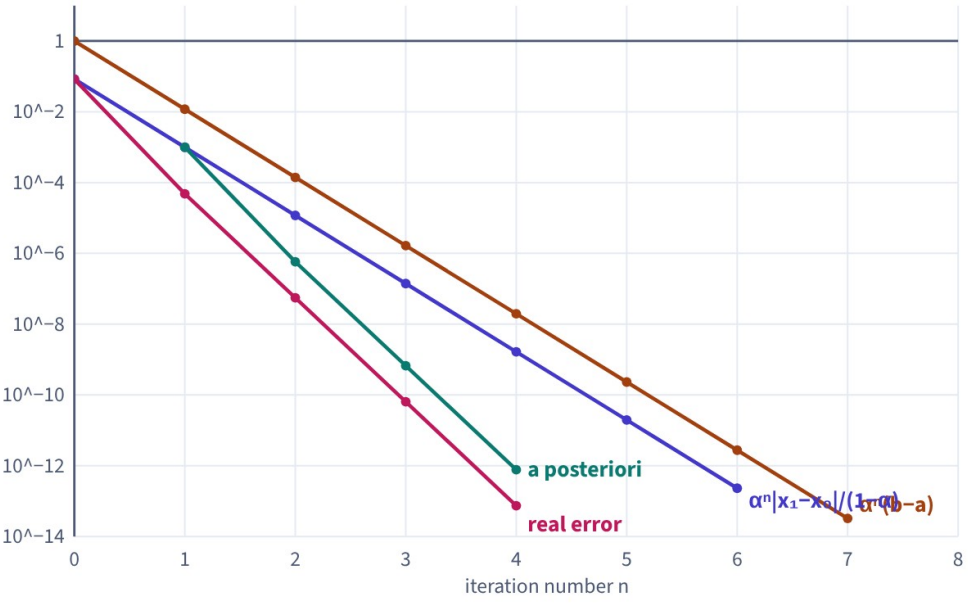
The estimate	What you need to know	Type
$\lvert x_n - p\rvert \le \alpha^n (b-a)$	α, a, b	a priori (the theorem)
$\lvert x_n - p\rvert \le \dfrac{\alpha^n}{1-\alpha}\lvert x_0 - x_1\rvert$	α, the first step	a priori (Remark 7)
$\lvert x_n - p\rvert \le \dfrac{\alpha}{1-\alpha}\lvert x_n - x_{n-1}\rvert$	α, the last step	a posteriori (addition)

Proof of the estimate in Remark 7

1. **Steps shrink geometrically.** $\lvert x_{k+1} - x_k\rvert = \lvert g(x_k) - g(x_{k-1})\rvert \le \alpha \lvert x_k - x_{k-1}\rvert$, so $\lvert x_{k+1} - x_k\rvert \le \alpha^k \lvert x_1 - x_0\rvert$.
2. **Distance over several steps.** With the triangle inequality, for $m>n$:
$$\lvert x_m - x_n\rvert \le \sum_{k=n}^{m-1} \lvert x_{k+1} - x_k\rvert \le \lvert x_1 - x_0\rvert \sum_{k=n}^{m-1} \alpha^k.$$
3. **Bound with the geometric series.** $\sum_{k=n}^{\infty} \alpha^k = \dfrac{\alpha^n}{1-\alpha}$, so $\lvert x_m - x_n\rvert \le \dfrac{\alpha^n}{1-\alpha}\lvert x_1 - x_0\rvert$.
4. **Pass to the limit.** For $m \to \infty$, $x_m \to p$ and we get $\lvert p - x_n\rvert \le \dfrac{\alpha^n}{1-\alpha}\lvert x_0 - x_1\rvert$. ■

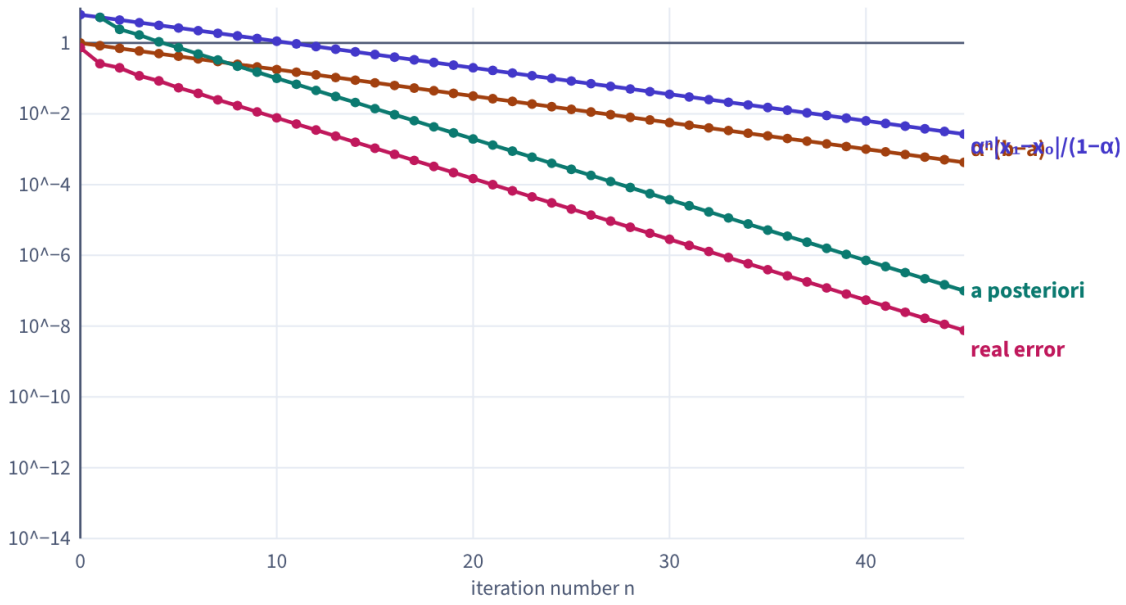
The a posteriori estimate follows by applying the same result with $n=1$, taking x_{n-1} as the new starting point: $\lvert x_n - p\rvert \le \dfrac{\alpha}{1-\alpha}\lvert x_{n-1} - x_n\rvert$.

The bounds compared with the real error



For ε = 10^{-4}, the a priori bound requires n = 2, while the real error drops below ε at n = 1. Bounds are guarantees, not exact predictions: they always stay above the real error.

The lecture example.



For ε = 10^{-4}, the a priori bound requires n = -1, while the real error drops below ε at n = 21. Bounds are guarantees, not exact predictions: they always stay above the real error.

x = cos x on [0, 1].

How to find the number of iterations

You want $\frac{\alpha^n}{1-\alpha}|x_1 - x_0| < \varepsilon$. Taking logarithms (careful, $\ln\alpha < 0$ flips the inequality):

$$n > \frac{\ln\frac{\varepsilon(1-\alpha)}{|x_1 - x_0|}}{\ln\alpha}.$$

For example, for $\cos x$ on $[0,1]$, with $x_0 = 0$, $\alpha = \sin 1$, $|x_1 - x_0| = 1$, $\varepsilon = 10^{-3}$, the formula gives $n \geq 51$. The table below shows that the real error drops below 10^{-3} already at $n = 16$: **the bound is a guarantee, not a forecast**.

n	x_n	$\lvert x_n - x_{n-1}\rvert$	bound $\alpha^n\lvert x_1 - x_0\rvert/(1-\alpha)$	real error
0	0.00000000	—	—	0.739085
1	1.00000000	1.000000	5.307994	0.260915
2	0.54030231	0.459698	4.466523	0.198783
3	0.85755322	0.317251	3.758449	0.118468
4	0.65428979	0.203263	3.162626	0.084795
5	0.79348036	0.139191	2.661258	0.054395
6	0.70136877	0.092112	2.239371	0.037716
7	0.76395968	0.062591	1.884366	0.024875
8	0.72210243	0.041857	1.585639	0.016983
9	0.75041776	0.028315	1.334269	0.011333

n	x_n	$\lvert x_n - x_{n-1}\rvert$	bound $\alpha^n\lvert x_1-x_0\rvert/(1-\alpha)$	real error
10	0.73140404	0.019014	1.122749	0.007681
11	0.74423735	0.012833	0.944761	0.005152
12	0.73560474	0.008633	0.794989	0.003480
13	0.74142509	0.005820	0.668960	0.002340
14	0.73750689	0.003918	0.562910	0.001578
15	0.74014734	0.002640	0.473673	0.001062
16	0.73836920	0.001778	0.398582	0.000716
17	0.73956720	0.001198	0.335395	0.000482
18	0.73876032	0.000807	0.282225	0.000325
19	0.73930389	0.000544	0.237484	0.000219
20	0.73893776	0.000366	0.199836	0.000147

A7. From f(x) = 0 to x = g(x)

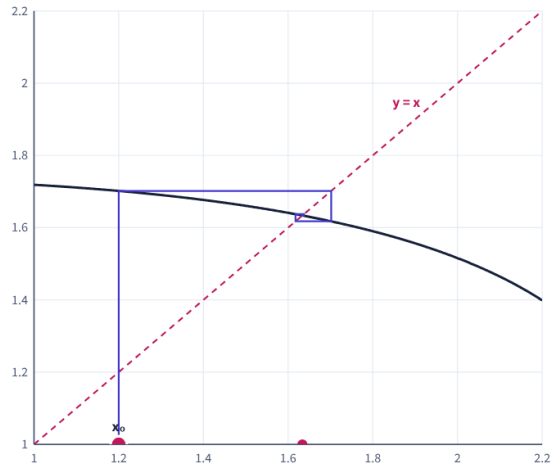
Level: intermediate

The theorem is about $x=g(x)$, but we usually have $f(x)=0$. We must rewrite the equation (Remark 2 in the source). An equation allows many rewritings, and **not all of them work**.

Take $x^5+x^3-16=0$, which has the unique root $\xi\approx 1.6337$ in $[1,2]$:

The rewriting $x=g(x)$	How we got it	$\lvert g'(\xi)\rvert$	Result
$x=\sqrt[5]{16-x^3}$	isolate x^5	≈ 0.22	fast
$x=\sqrt[3]{\dfrac{16}{1+x^2}}$	$x^3(x^2+1)=16$	≈ 0.48	slower
$x=\dfrac{16}{x^4+x^2}$	$x(x^4+x^2)=16$	≈ 3.5	diverges
$x=\sqrt[3]{16-x^5}$	isolate x^3	≈ 4.4	diverges

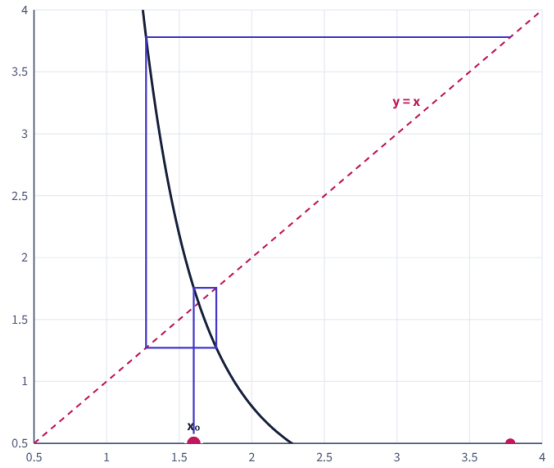
The same equation, four rewritings



n	x_n	$\lvert x_n - x_{n-1} \rvert$
0	1.20000000	–
1	1.70175473	0.50175473
2	1.61749675	0.08425798
3	1.63735044	0.01985369
4	1.63293703	0.00441340
5	1.63393159	0.00099456
6	1.63370815	0.00022344

After 6 steps: x_n = 1.63370815. From this x_0 the sequence settles at x ≈ 1.6337492. Good: $|g'| \leq 0.455$ on [1, 2].

The good rewriting: x = (16−x³)^(1/5).



n	x_n	$\lvert x_n - x_{n-1} \rvert$
0	1.60000000	–
1	1.75561798	0.15561798
2	1.27164607	0.48397191
3	3.78067949	2.50903343

After 3 steps: x_n = 3.78067949. From this x_0 the sequence does not settle: it oscillates or wanders. Bad: $|g'(\xi)| \approx 3.5$, the sequence runs away from the root.

The bad rewriting: x = 16/(x⁴+x²). The sequence runs away from the root.

The complete checking recipe

1. **Interval:** find $[a, b]$ containing a single root (Rolle's sequence, signs).
2. **Rewriting:** write $x = g(x)$, aiming for g to have a small slope near the root.
3. **Contraction:** compute $\alpha = \sup_{[a,b]} |g'(x)|$ and check $\alpha < 1$.
4. **Invariance:** check $g([a, b]) \subseteq [a, b]$. If g is monotonic, comparing $g(a)$ and $g(b)$ with a and b is enough.
5. **Iterations:** choose x_0, compute x_1 and determine n from the estimate.

Addition to the source

Remark 2 in the source mentions $g:[a,b] \to \mathbb{R}$ with $|g'| \leq \alpha < 1$. To apply the theorem, **the condition** $g([a,b]) \subseteq [a,b]$ is also needed. Counterexample: $g(x) = x/2 + 3$ on $[0,1]$ has $|g'| = 1/2$, but its fixed point is 6, outside the interval; a sequence started in $[0,1]$ leaves the interval immediately.

A8. Stopping, refinement and hypotheses

Level: advanced

When to stop (Remarks 5 and 6)

The iterative process continues until:

$$|x_{n+1} - x_n| \leq \varepsilon \text{ or } |f(x_n)| \leq \varepsilon .$$

If $x_n \neq 0$, the relative error can be used instead of the absolute error:

$$\left| 1 - \frac{x_{n+1}}{x_n} \right| \leq \varepsilon .$$

The relative error is suitable when solutions are very large or very small: an absolute $\varepsilon = 10^{-3}$ says nothing about a solution of the order of 10^{-6}.

In addition, **always limit the number of iterations** to some n_{max}. If you reach the limit without meeting the criterion, treat the situation as possible divergence, instead of getting stuck forever.

The trap of practical criteria

Problem	Stopping at	n	Criterion	Real error
$x = \sin x$, $x_0 = 1$ (root 0)	$\lvert x_{n+1} - x_n \rvert \leq \varepsilon$	87	0.000994	0.180440
$(x-1)^3 = 0$, Newton, $x_0 = 1.1$	$\lvert f(x_n) \rvert \leq \varepsilon$	1	0.000296	0.066667
$x = \cos x$, $x_0 = 0$	$\lvert x_{n+1} - x_n \rvert \leq \varepsilon$	18	0.000807	0.000325

The step dropped below ε, but the real error is still about 180·ε. The practical criterion is not a guarantee.

Important

A small step does not guarantee a small error, and a small value of $|f|$ does not guarantee closeness to the root. If you know α, the correct guarantee is $\frac{\alpha}{1-\alpha}|x_n - x_{n-1}| \leq \varepsilon$. When α is close to 1, the factor $\frac{\alpha}{1-\alpha}$ becomes very large.

Refining approximations (Remark 1)

The method can be used to **refine** initial approximations provided by more robust but slower methods, such as **bisection** or the **secant**. The strategy:

1. A few bisection steps trap the root in a small interval, on which g becomes a contraction.
2. Successive approximations or Newton quickly finish the job.

For $x^3+12x-1=0$ on $[0,1]$, four bisection steps give the interval $[0.0625, 0.125]$, which contains $\xi \approx 0.08329$. You implement this in L5.

The generalization (Remark 7)

The result remains valid for $g: M \to M$, g an α-contraction on $M \subseteq \mathbb{R}$, M **a closed set** (for example all of $\mathbb{R}$ or $[0,\infty)$). The estimate $\alpha^n(b-a)$ no longer makes sense on an unbounded set, but this one remains:

$$|x_n - \xi| \le \frac{\alpha^n}{1-\alpha}|x_0 - x_1|, \ \forall\, n \in \mathbb{N}^*.$$

Example. $g(x)=\frac{1+\sin x}{10}$ on $\mathbb{R}$: $|g'(x)|=\frac{|\cos x|}{10} \le \frac{1}{10}$, so the equation $10x-1=\sin x$ has exactly one solution on the whole real line.

Why all the hypotheses are needed

The missing hypothesis	Counterexample	What breaks
g maps the interval into itself	$g(x)=\frac{x}{2}+3$ on $[0,1]$	The fixed point, $p=6$, is outside the interval.
M closed	$g(x)=\frac{x}{2}$ on $(0,1]$	The sequence tends to 0, which is not in M: no fixed point.
A fixed, uniform $\alpha<1$	$g(x)=x+\frac{1}{x}$ on $[1,\infty)$	$\lvert g(x)-g(y)\rvert<\lvert x-y\rvert$, but always $g(x)>x$: no fixed point.
$\alpha<1$	$g(x)=x^2$ on $[0,1]$	Two fixed points (0 and 1): uniqueness is lost.
Continuity (Remark 3)	$g=1$ on $\left[0,\frac{1}{2}\right]$, $g=0$ on $(\frac{1}{2},1]$	The graph "jumps" over the diagonal: no fixed point.

A9. The lecture example, step by step

Level: intermediate

The problem. Compute with an error $\le 10^{-4}$ the roots of the equation $x^3+12x-1=0$.

1. **Isolating the roots.** $f(x)=x^3+12x-1$, $f'(x)=3x^2+12>0$. By Rolle's sequence (P3), there is a single real root. Since $f(0)=-1$ and $f(1)=12$, $\xi \in [0,1]$.
2. **The rewriting.** Factor out x: $x(x^2+12)=1$, so $x=g(x)$, where $g(x)=\frac{1}{x^2+12}$.
3. **Computing α.** $g'(x)=-\frac{2x}{(x^2+12)^2}$. The source directly writes $\alpha=\frac{2}{169}$; here is why. Let $u(x)=\frac{2x}{(x^2+12)^2}$. Differentiating, $u'(x)=\frac{2(12-3x^2)}{(x^2+12)^3}>0$ on $[0,1]$, so u is increasing and the maximum is at $x=1$:

$$\alpha = \sup_{[0,1]} |g'(x)| = u(1) = \frac{2}{13^2} = \frac{2}{169} \approx 0.0118.$$

4. **Invariance.** g is decreasing on $[0,1]$, with $g(0) = \frac{1}{12}$ and $g(1) = \frac{1}{13}$. So $g([0,1]) = \left[\frac{1}{13}, \frac{1}{12}\right] \subset [0,1]$. *The source does not write this check, but the theorem requires it.*

5. **The first iterations.** With $x_0 = 0$: $x_1 = g(0) = \frac{1}{12}$ and $x_2 = g(x_1) = \frac{1}{\frac{1}{144} + 12} = \frac{144}{1729} \approx 0.08328$.

6. **The number of iterations.** From the estimate, with $d(x_0, x_1) = |x_0 - x_1| = \frac{1}{12}$ and $1 - \alpha = \frac{167}{169}$:

$$\frac{1}{12} \cdot \frac{169}{167}\left(\frac{2}{169}\right)^n < 10^{-4} \Leftrightarrow \left(\frac{2}{169}\right)^n < \frac{167 \cdot 12}{169 \cdot 10^4} \approx 0.001186.$$

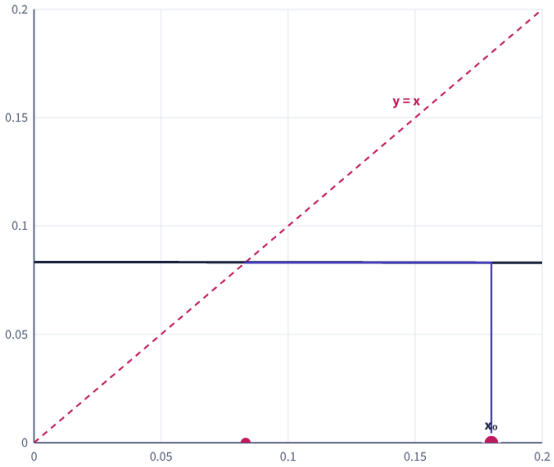
7. **Check the values.** For $n = 1$: $\frac{2}{169} \approx 0.0118 > 0.001186$: not enough. For $n = 2$: $\left(\frac{2}{169}\right)^2 \approx 0.00014 < 0.001186$: enough. So $n = 2$.

8. **Conclusion.** $\xi \approx x_2 \approx 0.08328$, with an accuracy better than 10^{-4}. ■

n	x_n	$\lvert x_n - x_{n-1}\rvert$	bound $\alpha^n \lvert x_1 - x_0\rvert/(1-\alpha)$	real error
0	0.00000000	—	—	0.083285
1	0.08333333	0.083333	0.000998	0.000048
2	**0.08328514**	**0.000048**	**0.000012**	**$5.56 \cdot 10^{-8}$**
3	0.08328519	$5.57 \cdot 10^{-8}$	$1.40 \cdot 10^{-7}$	$6.43 \cdot 10^{-11}$
4	0.08328519	$6.44 \cdot 10^{-11}$	$1.65 \cdot 10^{-9}$	$7.43 \cdot 10^{-14}$

At $n = 2$ the real error is about $6 \cdot 10^{-8}$, more than a thousand times smaller than required. The reason: α is very small, and the bound is taken over the whole interval $[0,1]$, even though the sequence stays near 0.083.

The example, in the diagram



n	x_n	$\lvert x_n - x_{n-1} \rvert$
0	0.18000000	–
1	0.08310894	0.09689106
2	0.08328539	0.00017646
3	0.08328519	2.04·10^-7

After 3 steps: $x_n = 0.08328519$. From this x_0 the sequence settles at $x \approx 0.0832852$. Almost zero slope, so very fast convergence.

The lecture example: the curve is almost horizontal, so the sequence stops almost immediately.

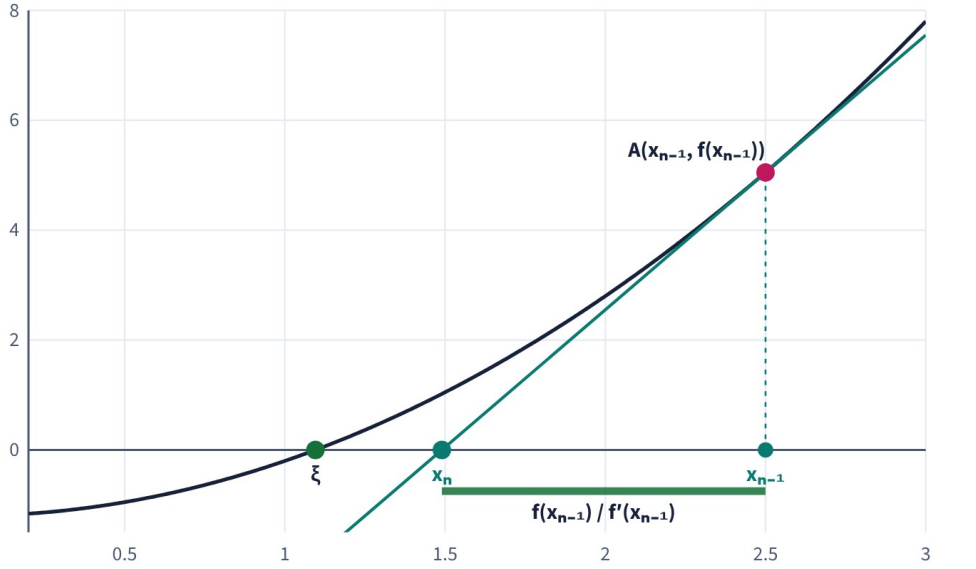
Lecture: Newton and the tangent

Replace the curve with its tangent and see where it crosses the axis. The number of correct decimals nearly doubles with every step.

B1. The geometric idea and the formula

Level: beginner

Let $f:[a,b]\to\mathbb{R}$ be of class $C^2([a,b])$, with the equation $f(x)=0$ having a unique solution $\xi\in[a,b]$. It is hard to find where a curve crosses the axis, but very easy to find where a **line** does. So we replace the curve with the tangent.



x_n is the intersection with Ox of the tangent to the graph of f at the point $A\left(x_{n-1},f\left(x_{n-1}\right)\right)$. This is the same drawing as in the source, redrawn.

Deriving the formula, in three lines

1. The equation of the tangent at $A\left(x_{n-1},f\left(x_{n-1}\right)\right)$ is: $y-f\left(x_{n-1}\right)=f'\left(x_{n-1}\right)\left(x-x_{n-1}\right)$.
2. Set $y=0$ and $x=x_n$: $-f\left(x_{n-1}\right)=f'\left(x_{n-1}\right)\left(x_n-x_{n-1}\right)$.
3. Divide by $f'\left(x_{n-1}\right)\neq 0$:

$$x_n=x_{n-1}-\frac{f\left(x_{n-1}\right)}{f'\left(x_{n-1}\right)},\ \forall\, n\in\mathbb{N}^*.$$

The first computation: square root of 2, by hand

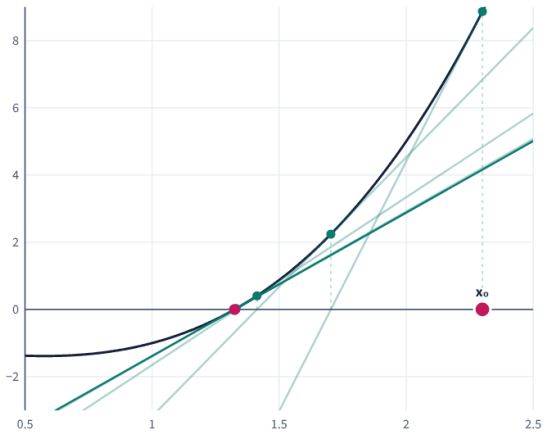
Let $f(x)=x^2-2$, $f'(x)=2x$, $x_0=2$. The formula becomes:

$$x_n=x_{n-1}-\frac{x_{n-1}^2-2}{2x_{n-1}}=\frac{1}{2}\left(x_{n-1}+\frac{2}{x_{n-1}}\right).$$

- $x_1 = \frac{1}{2}(2+1) = 1.5$
- $x_2 = \frac{1}{2}(1.5+1.3333) = 1.41666\ldots$
- $x_3 = 1.4142156\ldots$ (the exact value: $\sqrt{2} = 1.4142135\ldots$)

In three steps we have five correct decimals. This is the ancient Babylonian (Heron's) method for extracting square roots.

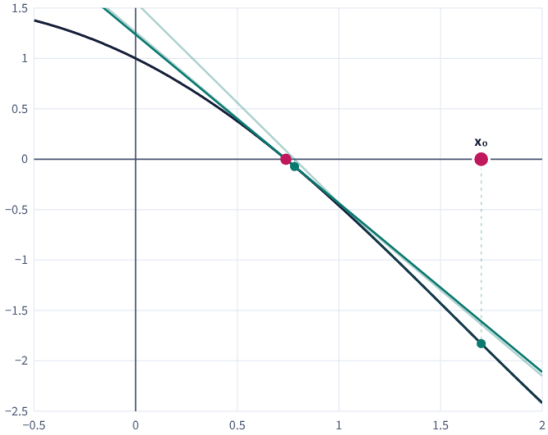
The tangent lab



n	x_n	$f(x_n)$	$\lvert\Delta\rvert$
0	2.3000000000	8.86700	—
1	1.7036987223	2.24144	0.59630
2	1.4128961166	0.40763	0.29080
3	1.3311867786	0.02775	0.08171
4	1.3247566124	0.00016	0.00643
5	1.3247179586	5.94·10^-9	3.87·10^-5

From this x_0 the method converges to $\xi \approx 1.32471796$. Now: n = 5, x_n = 1.32471796.

$f(x) = x^3 - x - 1$, from $x_0 = 2.3$.



n	x_n	$f(x_n)$	$\lvert\Delta\rvert$
0	1.7000000000	-1.82884	—
1	0.7817508625	-0.07207	0.91825
2	0.7394693815	-0.00064	0.04228
3	0.7390851658	-5.45·10^-8	0.00038
4	0.7390851332	-4.44·10^-16	3.26·10^-8

From this x_0 the method converges to $\xi \approx 0.73908513$. Now: n = 4, x_n = 0.73908513.

$f(x) = \cos x - x$, from $x_0 = 1.7$.

B2. Newton's theorem: local convergence

Level: advanced

Theorem (Newton's method)

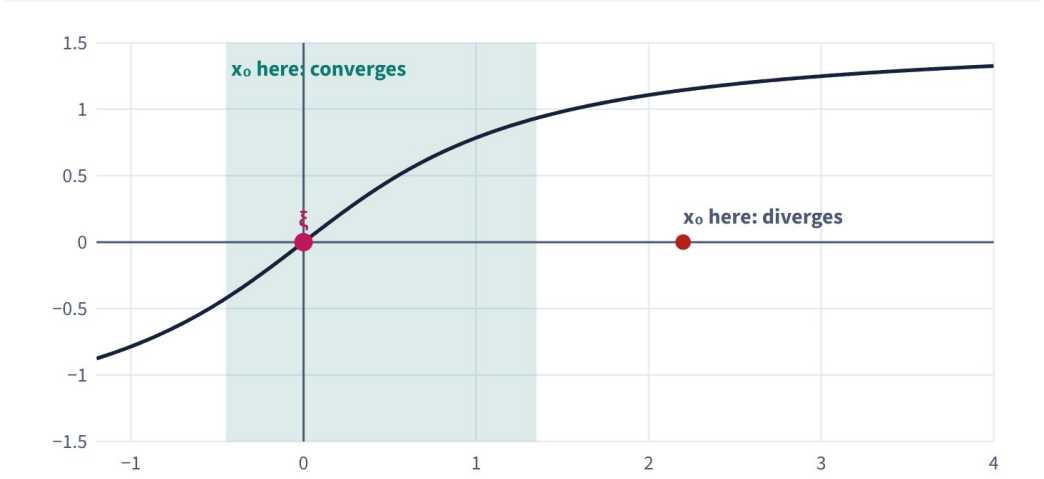
i) Under the above conditions, if in addition $f'(\xi)\neq 0$, then there exists $\delta>0$ such that, for every $x_0\in[\xi-\delta,\xi+\delta]$, the sequence $(x_n)_{n\in\mathbb{N}^*}$ given by $x_n=x_{n-1}-\dfrac{f(x_{n-1})}{f'(x_{n-1})}$ converges to ξ.

ii) If m_1, M_2 are given by $m_1=inf_{x\in[a,b]}|f'(x)|>0$ and $M_2=\sup_{x\in[a,b]}|f''(x)|$, then we have the estimates:

$$|\xi-x_n|\leq\frac{M_2}{2m_1}(x_n-x_{n-1})^2,\ \forall\, n\in\mathbb{N}^*,$$

$$|\xi-x_n|\leq\frac{M_2}{2m_1}(\xi-x_{n-1})^2,\ \forall\, n\in\mathbb{N}^*.$$

The word "local" is essential: the theorem guarantees convergence only **close enough** to the root.



For arctan x, there is a shaded zone around ξ in which the method converges. Far from the root, the method can diverge.

Proof of part i): Newton is a fixed-point method

This is where the two chapters meet. Newton is also an iteration $x_n=\varphi(x_{n-1})$, with:

$$\varphi(x)=x-\frac{f(x)}{f'(x)}.$$

1. Since f' is continuous and $f'(\xi)\neq 0$, f' does not vanish in a neighborhood of ξ, so φ is well defined and of class C^1 there.
2. Differentiate: $\varphi'(x)=1-\dfrac{f'(x)^2-f(x)f''(x)}{f'(x)^2}=\dfrac{f(x)f''(x)}{f'(x)^2}$.
3. At the root, $f(\xi)=0$, so $\varphi'(\xi)=0$. *This is the essence of the speed: the slope of the iteration is zero exactly at the root.*
4. By continuity, there is $\delta>0$ with $|\varphi'(x)|\leq\dfrac{1}{2}$ on $I=[\xi-\delta,\xi+\delta]$. So φ is a $\dfrac{1}{2}$-contraction on I (Remark 2).
5. φ maps I into I: $|\varphi(x)-\xi|=|\varphi(x)-\varphi(\xi)|\leq\dfrac{1}{2}|x-\xi|\leq\delta$.
6. Apply the theorem from A5: for every $x_0\in I$, $x_n\to$ the unique fixed point of φ in I, that is ξ. ■

Proof of part ii)

We assume that the terms of the sequence stay in $[a, b]$.

1. **The first inequality.** Write Taylor around x_{n-1}, evaluated at x_n:
$$f(x_n) = f(x_{n-1}) + f'(x_{n-1})(x_n - x_{n-1}) + \frac{f''(c)}{2}(x_n - x_{n-1})^2.$$
2. From Newton's formula, $f(x_{n-1}) + f'(x_{n-1})(x_n - x_{n-1}) = 0$. What remains is $f(x_n) = \frac{f''(c)}{2}(x_n - x_{n-1})^2$, so $|f(x_n)| \le \frac{M_2}{2}(x_n - x_{n-1})^2$.
3. From Lagrange, $|f(x_n)| = |f(x_n) - f(\xi)| = |f'(d)| \cdot |x_n - \xi| \ge m_1 |x_n - \xi|$.
4. Combining: $m_1 |\xi - x_n| \le \frac{M_2}{2}(x_n - x_{n-1})^2$, as required. ■
5. **The second inequality.** Write Taylor around x_{n-1}, evaluated at ξ:
$$0 = f(\xi) = f(x_{n-1}) + f'(x_{n-1})(\xi - x_{n-1}) + \frac{f''(c)}{2}(\xi - x_{n-1})^2.$$
6. Divide by $f'(x_{n-1})$ and use $x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}$: we get $\xi - x_n = -\frac{f''(c)}{2f'(x_{n-1})}(\xi - x_{n-1})^2$. In absolute value, this gives the second estimate. ■

B3. Quadratic convergence: digits double

Level: intermediate

The second estimate says that the new error is proportional to the **square** of the old error. Writing $K = \frac{M_2}{2m_1}$ and $e_n = |\xi - x_n|$:

$$e_n \le K\, e_{n-1}^2.$$

If $e_{n-1} \approx 10^{-3}$ and $K \approx 1$, then $e_n \approx 10^{-6}$, then 10^{-12}. The number of correct decimals nearly doubles at each step. Here is Newton for $\sqrt{2}$, computed with 130 digits (the correct decimals are marked in green):

x_0 2,0000000000000… (0 correct decimals)

x_1 1,5000000000000… (0 correct decimals)

x_2 1,41666666666666… (2 correct decimals)

x_3 1,41421356862745… (5 correct decimals)

x_4 1,41421356237346899106… (11 correct decimals)

x_5 1,41421356237309504880168962350 25… (23 correct decimals)

x_6 1,41421356237309504880168872420969807856967187537 72340015… (47 correct decimals)

x_7
1,41421356237309504880168872420969807856967187537694807317667973799073247846210703885038753432764160163978… (96 correct decimals)

x_8
1,41421356237309504880168872420969807856967187537694807317667973799073247846210703885038753432764157273501384623 0… (110+ correct decimals)

x_9
1,41421356237309504880168872420969807856967187537694807317667973799073247846210703885038753432764157273501384623 0… (110+ correct decimals)

Successive approximations	*Newton*
$e_n \le \alpha\, e_{n-1}$ **Linear** convergence: you gain a fixed number of digits at each step.	$e_n \le K\, e_{n-1}^2$ **Quadratic** convergence: you double the number of digits at each step.

The first estimate, with $\left(x_n - x_{n-1}\right)^2$, is of great practical value, because ξ does not appear in it. You compute it directly and it gives you a **stopping criterion with a guarantee**:

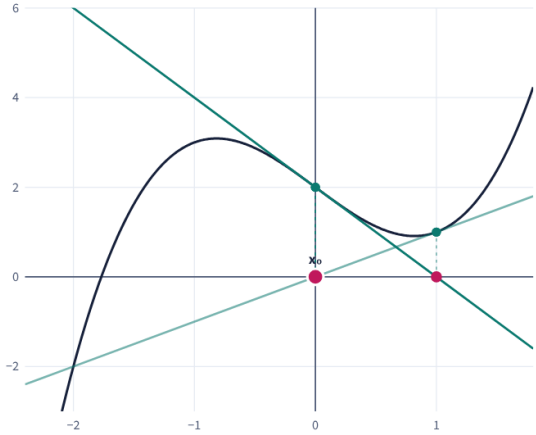
$$\frac{M_2}{2\,m_1}\left(x_n - x_{n-1}\right)^2 \le \varepsilon \quad \Rightarrow \quad \left|\xi - x_n\right| \le \varepsilon\,.$$

B4. When Newton fails

Level: intermediate

The source names the main disadvantage: x_0 must be in $\left[\xi - \delta\,, \xi + \delta\right]$, but **neither ξ nor δ is known a priori**. Here is what can happen:

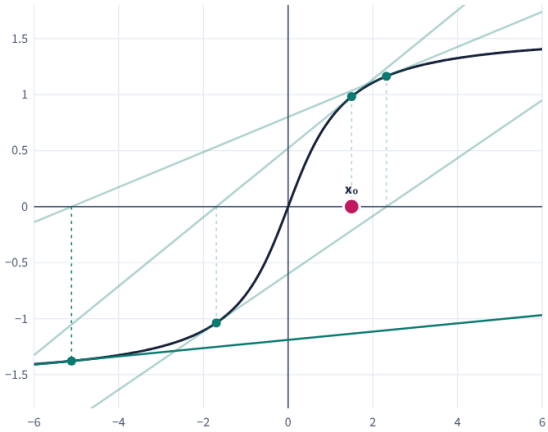
The gallery of failures



n	x_n	$f(x_n)$	$\lvert\Delta\rvert$
0	0.0000000000	2.00000	–
1	1.0000000000	1.00000	1.00000
2	0.0000000000	2.00000	1.00000
3	1.0000000000	1.00000	1.00000
4	0.0000000000	2.00000	1.00000
5	1.0000000000	1.00000	1.00000

From this x_0 the method does not reach a root in 60 steps. Now: n = 5, x_n = 1.00000000.

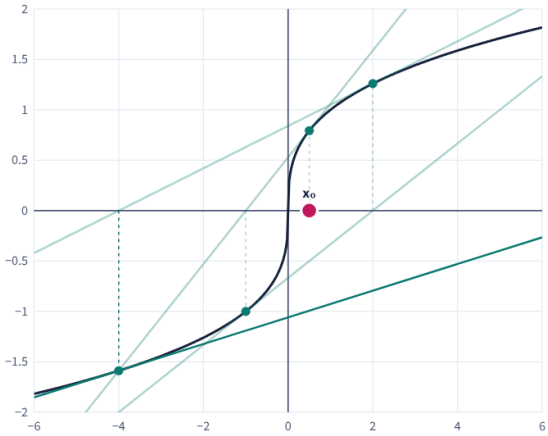
Cycle: $f(x) = x^3 - 2x + 2$, from $x_0 = 0$.



n	x_n	$f(x_n)$	$\lvert\Delta\rvert$
0	1.5000000000	0.98279	–
1	-1.6940796006	-1.03755	3.19408
2	2.3211269614	1.16400	4.01521
3	-5.1140878368	-1.37769	7.43521
4	32.2956839142	1.53984	37.40977

From this x_0 the method gets stuck or blows up. Now: n = 4, x_n = 32.29568391.

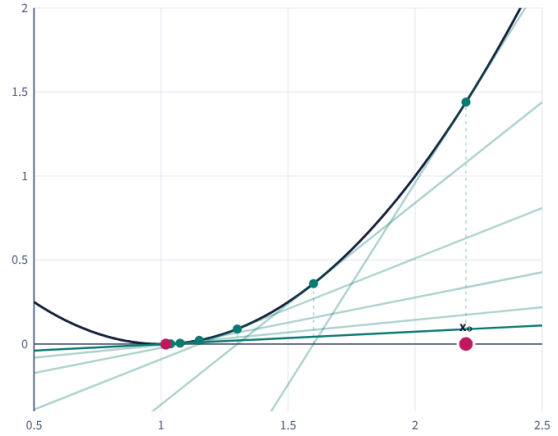
Overshoot: $f(x) = \arctan x$, from $x_0 = 1.5$.



n	x_n	$f(x_n)$	$\lvert\Delta\rvert$
0	0.5000000000	0.79370	—
1	-1.0000000000	-1.00000	1.50000
2	2.0000000000	1.25992	3.00000
3	-4.0000000000	-1.58740	6.00000
4	8.0000000000	2.00000	12.00000

From this x_0 the method gets stuck or blows up. Now: n = 4, x_n = 8.00000000.

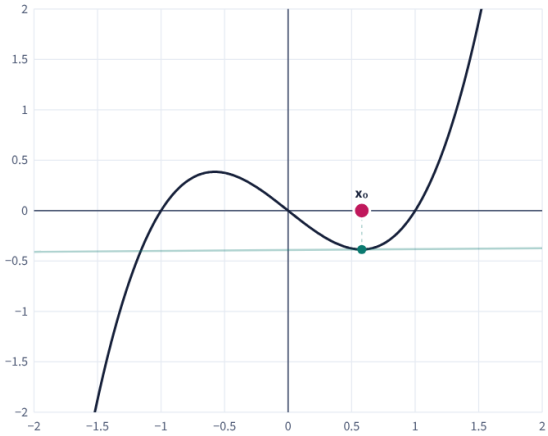
Moving away: $f(x) = \sqrt[3]{x}$, from $x_0 = 0.5$.



n	x_n	$f(x_n)$	$\lvert\Delta\rvert$
0	2.2000000000	1.44000	—
1	1.6000000000	0.36000	0.60000
2	1.3000000000	0.09000	0.30000
3	1.1500000000	0.02250	0.15000
4	1.0750000000	0.00562	0.07500
5	1.0375000000	0.00141	0.03750
6	1.0187500000	0.00035	0.01875

From this x_0 the method converges to $\xi \approx 1.00000000$. Now: n = 6, x_n = 1.01875000.

Only linear convergence: $f(x) = (x-1)^2$, from $x_0 = 2.2$.



n	x_n	$f(x_n)$	$\lvert\Delta\rvert$
0	0.5800000000	-0.38489	–
1	42.4156521739	76267.05607	41.83565
2	28.2823415762	22594.50376	14.13331
3	18.8627549379	6692.57192	9.41959

From this x_0 the method converges to $\xi \approx 1.00000000$. Now: n = 3, x_n = 18.86275494.

Nearly horizontal tangent: f(x) = x³–x, from x_0 = 0.58.

The situation	Example	What happens
Cycle	x^3-2x+2, $x_0=0$	$x_1=1$, $x_2=0$, $x_3=1,\ldots$ forever.
Overshoot	$\arctan x$, $x_0=1.5$	$\lvert x_n\rvert$ grows. The threshold is $\lvert x_0\rvert\approx 1.3917$.
Moving away	$\sqrt[3]{x}$, any $x_0\neq 0$	$x_n=-2x_{n-1}$: it doubles. $f'(0)$ does not exist.
Multiple root	$(x-1)^2$, $f'(\xi)=0$	It converges, but only linearly: $e_n=\frac{1}{2}e_{n-1}$.
Horizontal tangent	x^3-x, near $\pm\frac{1}{\sqrt{3}}$	$f'\approx 0$: the next point jumps very far.

Basins of attraction: where you end up from each x_0

For $f(x)=x^3-x$, with roots -1, 0, 1, each x_0 leads to a certain root. The next band colors the axis by destination. Near $x\approx 0.447$ (and symmetrically), the destination changes chaotically.



Displayed interval: [−1.55000, 2.45000]. Touch the band to choose an x_0.

Basins for x³–x: blue reaches –1, yellow reaches 0, green reaches 1.

The message

The root Newton reaches is not necessarily the one closest to x_0. That is why we need a safe criterion for choosing x_0: the tangent theorem.

B5. The tangent theorem: choosing x_0 safely

Level: intermediate

Theorem (tangent method)

Let $f:[a,b]\to\mathbb{R}$ be twice differentiable such that $f(a)\cdot f(b)<0$, $f'(x)\neq 0$ and $f''(x)\neq 0$, for every $x\in[a,b]$. Then the sequence given by $x_0\in[a,b]$ with

$$f(x_0)\cdot f''(x_0)>0,\ x_n=x_{n-1}-\frac{f(x_{n-1})}{f'(x_{n-1})},\ \forall\, n\in\mathbb{N}^*,$$

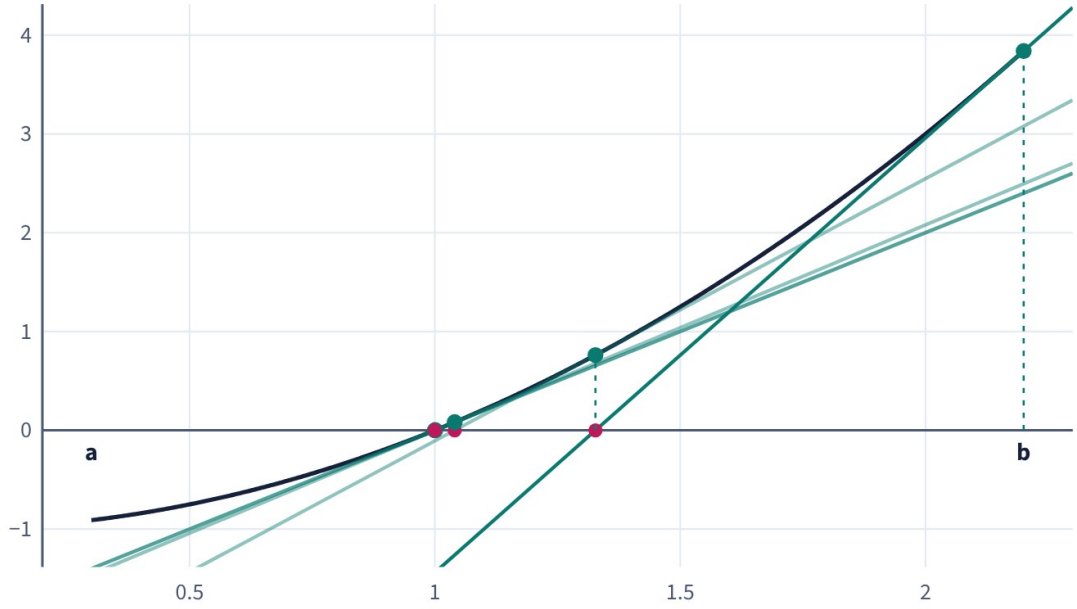
converges strictly monotonically to ξ, the unique solution of the equation $f(x)=0$.

What each hypothesis says

The condition	Meaning
$f(a)f(b)<0$	A root exists (the function changes sign).
$f'\neq 0$	f is strictly monotonic, so the root is unique. Tangents are not horizontal.
$f''\neq 0$	The curve has constant concavity, it does not "twist".
$f(x_0)f''(x_0)>0$	We start on the side where the curve lies above the tangent, toward the root.

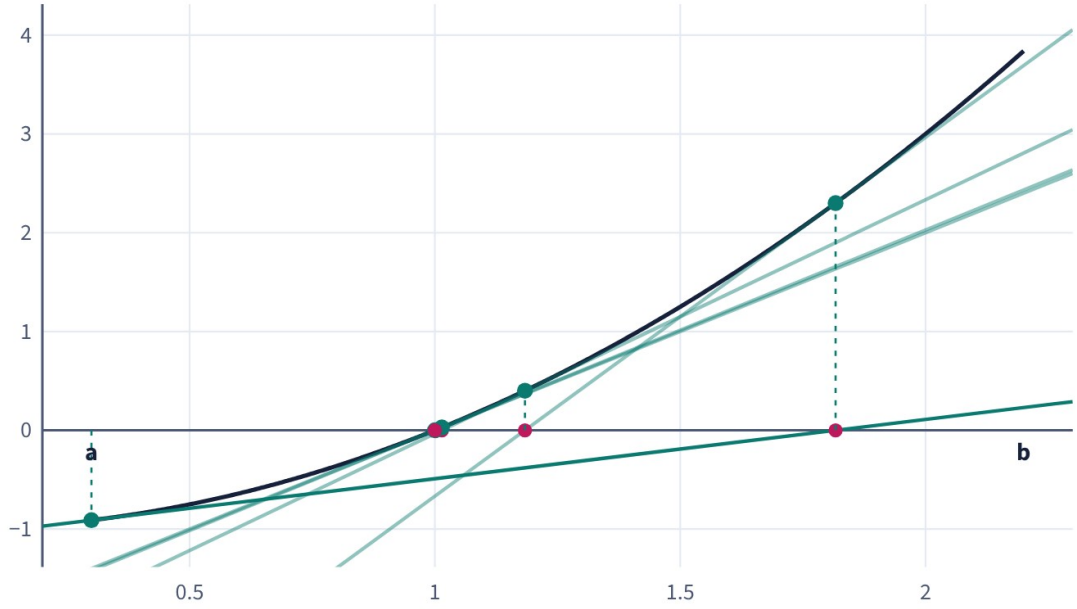
Derivatives cannot change sign without vanishing (Darboux's property). So f' and f'' have constant sign, and the graph falls into one of **four cases**.

The four cases



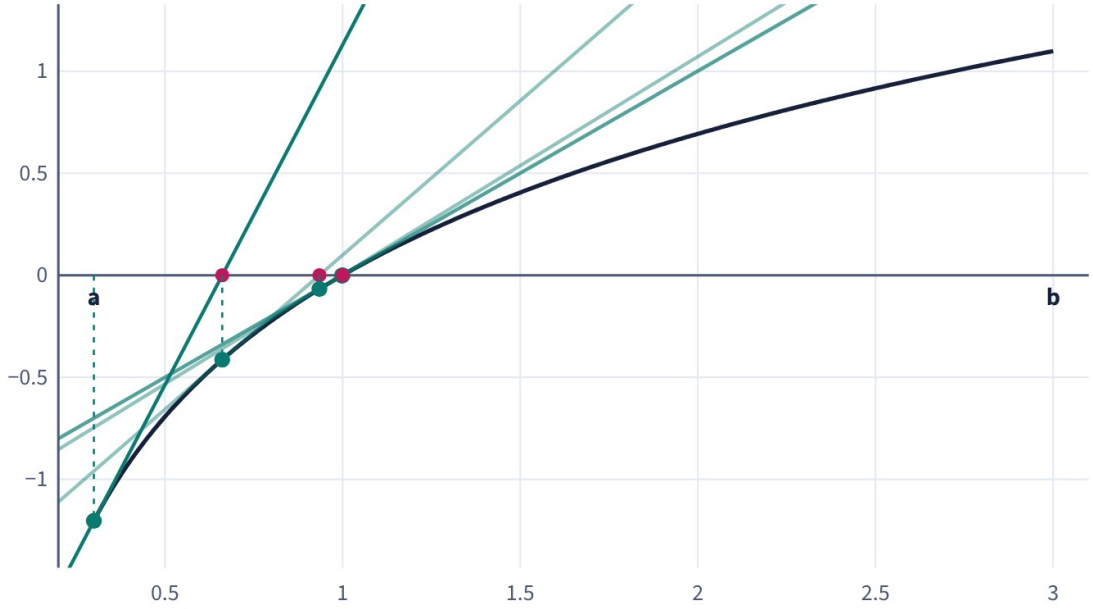
$f(x_0)\cdot f''(x_0) > 0$. The condition in the theorem is satisfied: the sequence strictly decreases toward ξ, without leaving the interval. Sequence: 2.2000, 1.3273, 1.0403, 1.0008, 1.0000.

f'>0, f''>0: correct start from b.



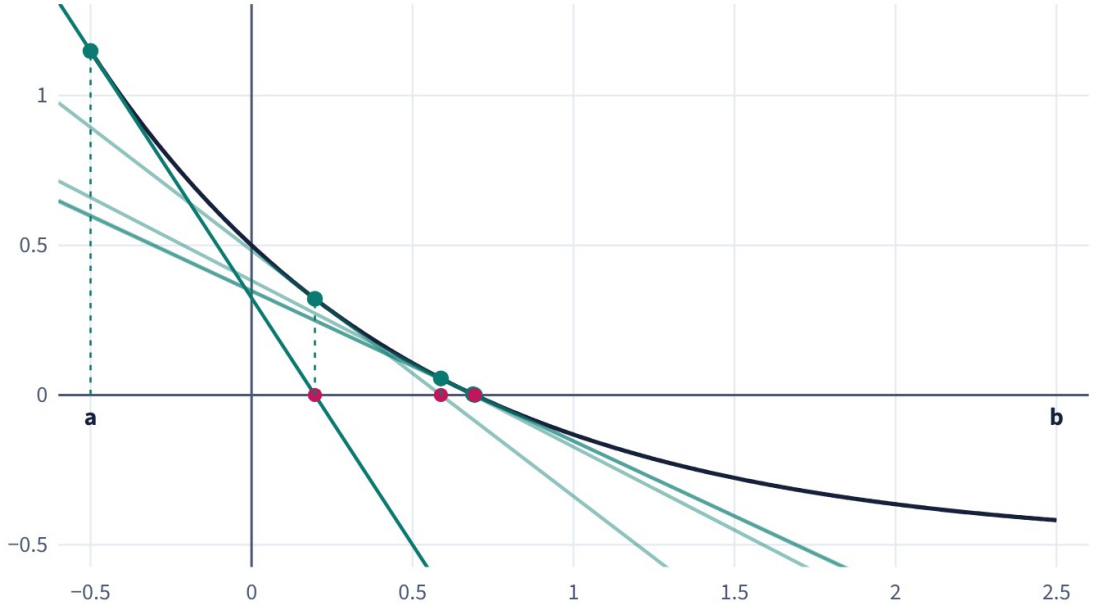
$f(x_0)\cdot f''(x_0) < 0$. The condition is not satisfied: the first step jumps to the other side of the root, so monotonicity from x_0 is lost. Sequence: 0.3000, 1.8167, 1.1836, 1.0142, 1.0001.

f'>0, f''>0: start from a, the condition is not satisfied.



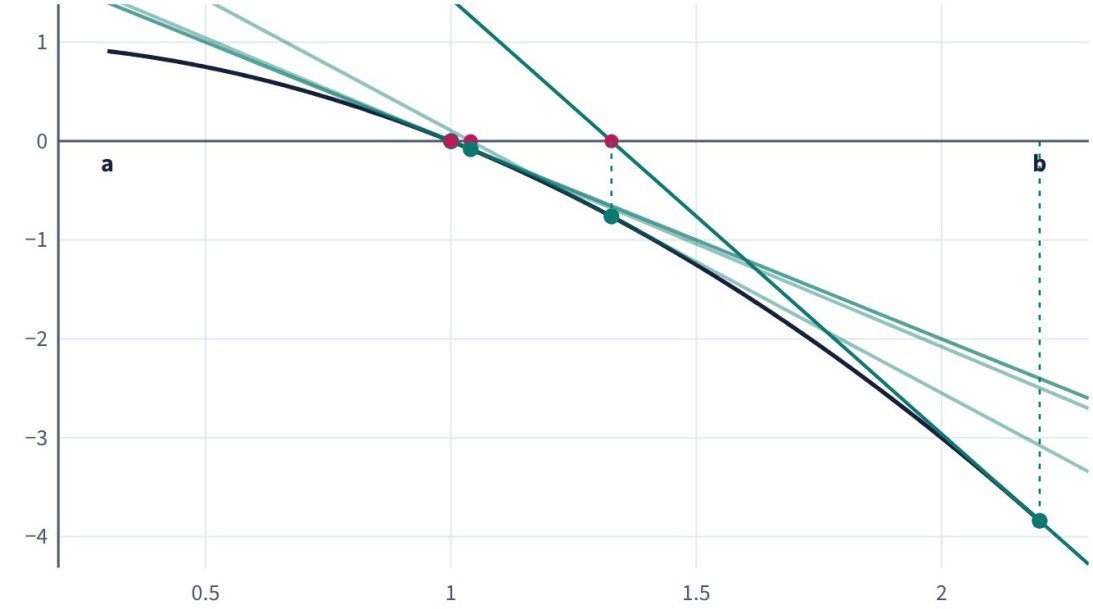
$f(x_0)\cdot f''(x_0) > 0$. The condition in the theorem is satisfied: the sequence strictly increases toward ξ, without leaving the interval. Sequence: 0.3000, 0.6612, 0.9347, 0.9978, 1.0000.

f'>0, f''<0: correct start from a.



$f(x_0)\cdot f''(x_0) > 0$. The condition in the theorem is satisfied: the sequence strictly increases toward ξ, without leaving the interval. Sequence: −0.5000, 0.1967, 0.5880, 0.6878, 0.6931.

f'<0, f">0: correct start from a.



$f(x_0)\cdot f''(x_0) > 0$. The condition in the theorem is satisfied: the sequence strictly decreases toward ξ, without leaving the interval. Sequence: 2.2000, 1.3273, 1.0403, 1.0008, 1.0000.

f'<0, f"<0: correct start from b.

Signs	Shape	x_0	The sequence
$f'>0, f''>0$	increasing, convex	b	decreasing

Signs	Shape	x_0	The sequence
$f'>0, f''<0$	increasing, concave	a	increasing
$f'<0, f''>0$	decreasing, convex	a	increasing
$f'<0, f''<0$	decreasing, concave	b	decreasing

The short rule

Choose the endpoint where f has **the same sign as** f''. You do not need the table: compute $f(a)f''(a)$ and $f(b)f''(b)$ and take the positive one.

The proof (case $f'>0, f''>0$)

The other cases reduce to this one, by replacing f with $-f$ or x with $-x$.

1. **Uniqueness.** f is strictly increasing and changes sign, so it has exactly one root ξ. The condition $f(x_0)f''(x_0)>0$ with $f''>0$ gives $f(x_0)>0$, so $x_0>\xi$.
2. **Induction.** Suppose $x_{n-1}>\xi$. Then $f(x_{n-1})>0$ and $f'(x_{n-1})>0$, so $x_n = x_{n-1} - \dfrac{f(x_{n-1})}{f'(x_{n-1})} < x_{n-1}$: the sequence strictly decreases.
3. **It does not pass the root.** From B2, $x_n - \xi = \dfrac{f''(c)}{2f'(x_{n-1})}(\xi - x_{n-1})^2 > 0$, so $x_n>\xi$. Thus $\xi<x_n<x_{n-1}\le x_0$ and we stay in $[a,b]$.
4. **A limit exists.** The sequence is decreasing and bounded below, so $x_n \to L \in [\xi, x_0]$.
5. **The limit is the root.** f' and f are continuous (f is twice differentiable). From $f'(x_{n-1})(x_{n-1}-x_n)=f(x_{n-1})$, in the limit $f'(L)\cdot 0 = f(L)$, so $f(L)=0$ and $L=\xi$. ■

Trap: f" vanishes at an endpoint

For $f(x)=x^3+12x-1$ on $[0,1]$, $f''(x)=6x$ vanishes at 0, so the hypothesis "$f''\neq 0$ on $[a,b]$" is **not satisfied** on $[0,1]$. Solution: restrict the interval to $[0.05,1]$, where $f(0.05)<0$, $f(1)>0$, $f''>0$. The complete solution is in S3.

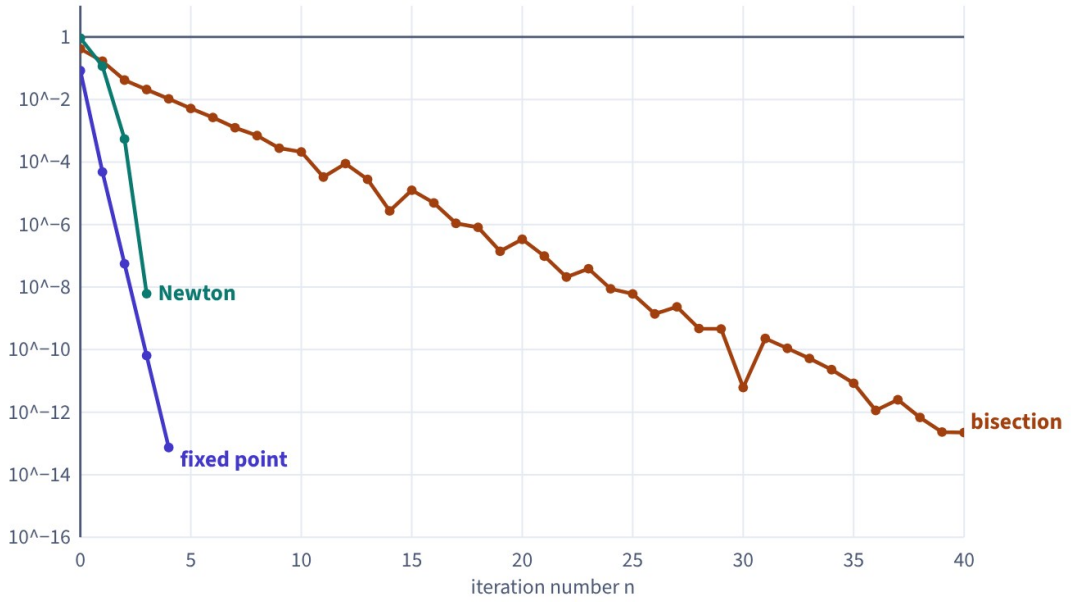
Monotonicity also gives you a test

If, under the conditions of the theorem, the sequence is not monotonic, you have a computation or hypothesis error. It is a quick check, useful in an exam.

B6. Comparing the methods

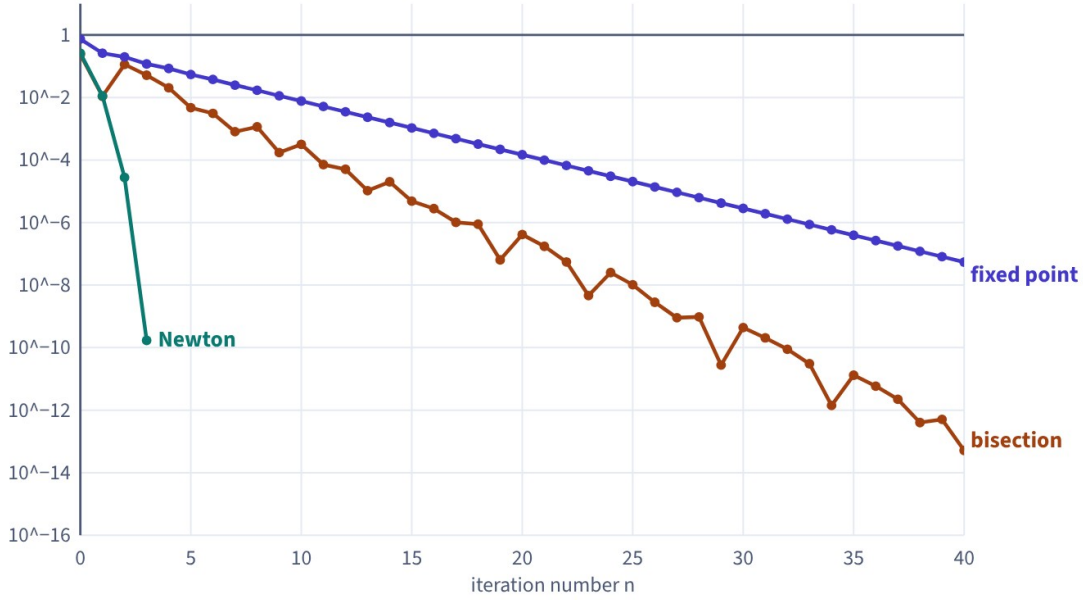
Level: intermediate

The race of the methods



Iterations until error < 10^{-10}: bisection 30, fixed point 3, Newton 4. A Newton step, however, costs one evaluation of f and one of f′.

$x^3+12x-1 = 0$ on [0, 1].



Iterations until error < 10^{-10}: bisection 29, fixed point >40, Newton 4. A Newton step, however, costs one evaluation of f and one of f′.

cos x − x = 0 on [0, 1].

A special case: $g(x)=\frac{1}{x}+\frac{x}{2}$ is exactly the Newton iteration for x^2-2, so the two methods coincide.

	Bisection	Successive approximations	Newton / tangent
Formula	$c=\frac{a+b}{2}$	$x_{n+1}=g\left(x_n\right)$	$x_n=x_{n-1}-\frac{f\left(x_{n-1}\right)}{f'\left(x_{n-1}\right)}$
Requires	a sign change	a rewriting; g a contraction	f'; $f'\neq 0$; a good x_0
Speed	linear, factor $\frac{1}{2}$	linear, factor α	quadratic (simple root)
Guarantee	always, if f is continuous	if the hypotheses are satisfied	local; global with the tangent theorem
Cost per step	one evaluation of f	one evaluation of g	one evaluation of f and one of f'
Error	$\frac{b-a}{2^{n+1}}$	$\frac{\alpha^n}{1-\alpha}\left\|x_1-x_0\right\|$	$\frac{M_2}{2m_1}\left(x_n-x_{n-1}\right)^2$

How to choose

If the derivative is easy to compute, use Newton with a correctly chosen x_0. If you do not have f' or you already have a clear contraction, use successive approximations. If you know nothing, start with bisection and then refine.

Seminar

From recipes applied step by step to problems and proofs. Try on your own first, then read the solution.

S1. The worksheet and the checker

Level: beginner

Successive approximations	*Tangent method*
1. Isolate the root: $[a,b]$.	1. $f(a)f(b)<0$?
2. Rewrite: $x=g(x)$.	2. $f'\neq 0$ on $[a,b]$?
3. $\alpha=\sup\lvert g'\rvert<1$?	3. $f''\neq 0$ on $[a,b]$?
4. $g([a,b])\subseteq[a,b]$?	4. Choose the endpoint with $f(x_0)f''(x_0)>0$.
5. Choose x_0, compute x_1.	5. Compute m_1, M_2, $K=\frac{M_2}{2m_1}$.
6. $\frac{\alpha^n}{1-\alpha}\lvert x_1-x_0\rvert<\varepsilon$: find n.	6. Iterate Newton.
7. Compute up to x_n.	7. Stop when $K(x_n-x_{n-1})^2\le\varepsilon$.

The hypothesis checker

✓ f is defined on the whole interval

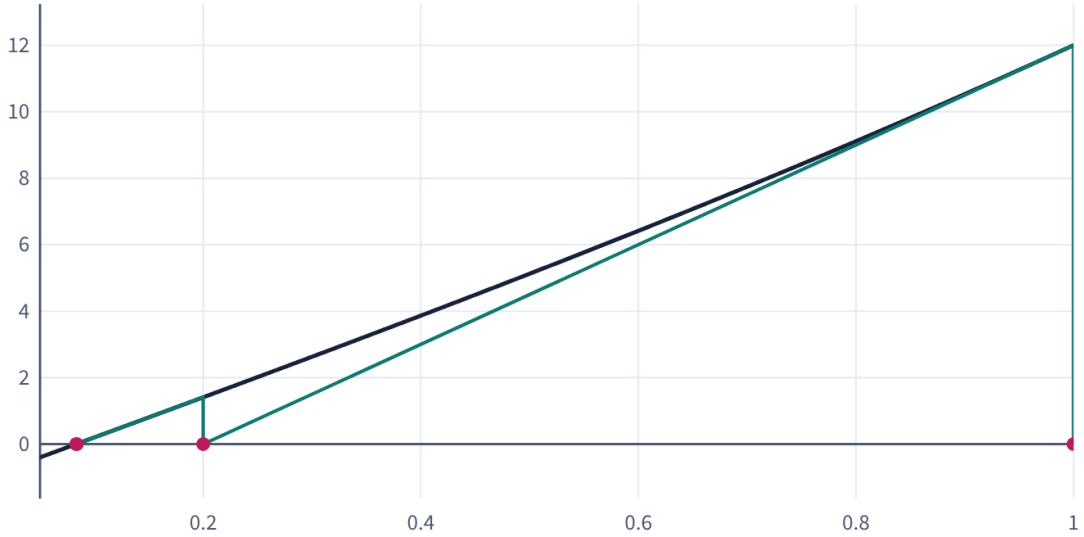
✓ f(a)·f(b) < 0: f(a) = −0.3999, f(b) = 12.0000

✓ f′ does not vanish: f′ takes values between 12.0075 and 15.0000

✓ f″ does not vanish: f″ takes values between 0.3000 and 6.0000

✓ The endpoint with $f(x_0)\cdot f''(x_0) > 0$ is $x_0 = 1.0000$

$m_1 \approx 12.0075$, $M_2 \approx 6.0000$, $M_2/(2m_1) \approx 0.2498$. Stop at n = 3: $\xi \approx$ **0.0832851977**, with error ≤ 7.37·10^−8. The check is numerical: it helps, but it does not replace a proof.



n	x_n	$f(x_n)$	$M_2/(2m_1)\cdot\Delta^2$
0	1.0000000000	12.0000	–
1	0.2000000000	1.4080	0.1599
2	0.0838283828	0.0065	0.0034
3	0.0832851977	7.40·10^-8	7.37·10^-8

Checking the hypotheses of the tangent method: $x^3+12x-1$ on [0.05, 1].

S2. The four exercises from the lecture, solved

Level: intermediate

Statement: *using the contraction principle, solve the equations below with an approximation of* 10^{-3}. The bold row in the tables is the value required by the a priori bound.

Exercise 1. $x^3+4x-1=0$

Complete solution

1. **Isolation.** $f'(x)=3x^2+4>0$, so there is a single root. $f(0)=-1$, $f(1)=4$, so $\xi\in[0,1]$.
2. **Rewriting.** $x(x^2+4)=1$, that is $x=g(x)=\dfrac{1}{x^2+4}$.
3. **α.** $|g'(x)|=u(x)=\dfrac{2x}{(x^2+4)^2}$; $u'(x)=\dfrac{2(4-3x^2)}{(x^2+4)^3}>0$ on $[0,1]$. So $\alpha=u(1)=\dfrac{2}{25}=0.08$.
4. **Invariance.** g is decreasing: $g([0,1])=\left[\dfrac{1}{5},\dfrac{1}{4}\right]\subset[0,1]$.
5. **Iterations.** $x_0=0$, $x_1=\dfrac{1}{4}$, so $\dfrac{0.25}{0.92}\cdot 0.08^n<10^{-3}$. $n=2$: $1.74\cdot 10^{-3}$, not enough. $n=3$: $1.39\cdot 10^{-4}$, enough.
6. **Result.** $\xi\approx x_3\approx 0.2463$.

n	x_n	$\lvert x_n - x_{n-1}\rvert$	bound $\alpha^n\lvert x_1-x_0\rvert/(1-\alpha)$	real error
0	0.00000000	—	—	0.246266
1	0.25000000	0.250000	0.021739	0.003734
2	0.24615385	0.003846	0.001739	0.000112
3	**0.24626953**	**0.000116**	**0.000139**	**$3.35\cdot10^{-6}$**
4	0.24626607	$3.45\cdot10^{-6}$	0.000011	$1.00\cdot10^{-7}$
5	0.24626618	$1.03\cdot10^{-7}$	$8.90\cdot10^{-7}$	$2.99\cdot10^{-9}$

Exercise 2. $10x-1=\sin x$

Complete solution

1. **Rewriting.** $x=g(x)=\dfrac{1+\sin x}{10}$, defined on $M=\mathbb{R}$, a closed set.
2. **α.** $|g'(x)|=\dfrac{|\cos x|}{10}\le\dfrac{1}{10}$, so $\alpha=0.1$ on the whole axis. Remark 7 applies: there is exactly one real solution.
3. **Iterations.** $x_0=0$, $x_1=0.1$, so $\dfrac{0.1}{0.9}\cdot 0.1^n<10^{-3}$. $n=2$: $1.11\cdot 10^{-3}$, not enough. $n=3$: $1.11\cdot 10^{-4}$, enough.
4. **Result.** $\xi\approx x_3\approx 0.1110$.

n	x_n	$\lvert x_n - x_{n-1}\rvert$	bound $\alpha^n\lvert x_1-x_0\rvert/(1-\alpha)$	real error
0	0.00000000	—	—	0.111086
1	0.10000000	0.100000	0.011111	0.011086
2	0.10998334	0.009983	0.001111	0.001102
3	**0.11097617**	**0.000993**	**0.000111**	**0.000110**
4	0.11107485	0.000099	0.000011	0.000011
5	0.11108466	$9.81\cdot10^{-6}$	$1.11\cdot10^{-6}$	$1.08\cdot10^{-6}$

Exercise 3. $x=\dfrac{1}{3}(1-e^{-x})$

Complete solution

The form $x = g(x)$ is already given, which tempts you to iterate directly. But how many solutions are there?

1. **Isolation.** Let $h(x) = x - \frac{1}{3}\left(1 - e^{-x}\right)$. $h'(x) = 1 - \frac{1}{3}e^{-x} = 0$ gives $x = -\ln 3 \approx -1.0986$, where $h(-\ln 3) = -\ln 3 + \frac{2}{3} \approx -0.432 < 0$. The limits at $\pm\infty$ are $+\infty$. So **there are two roots**: one less than $-\ln 3$, the other greater.
2. **The obvious root.** $x = 0$: $\frac{1}{3}(1-1) = 0$. ✓. On $[0,1]$: $g'(x) = \frac{1}{3}e^{-x} \in \left(0, \frac{1}{3}\right]$, $\alpha = \frac{1}{3}$, and $g([0,1]) = [0, 0.2107] \subset [0,1]$. From $x_0 = 1$: $\frac{0.7893}{2/3} \cdot 3^{-n} < 10^{-3}$ gives $n = 7$.

n	x_n	$\lvert x_n - x_{n-1}\rvert$	bound $\alpha^n \lvert x_1 - x_0\rvert/(1-\alpha)$	real error
0	1.00000000	—	—	1.000000
1	0.21070685	0.789293	0.394647	0.210707
2	0.06332951	0.147377	0.131549	0.063330
3	0.02045529	0.042874	0.043850	0.020455
4	0.00674917	0.013706	0.014617	0.006749
5	0.00224215	0.004507	0.004872	0.002242
6	0.00074655	0.001496	0.001624	0.000747
7	**0.00024876**	**0.000498**	**0.000541**	**0.000249**
8	0.00008291	0.000166	0.000180	0.000083

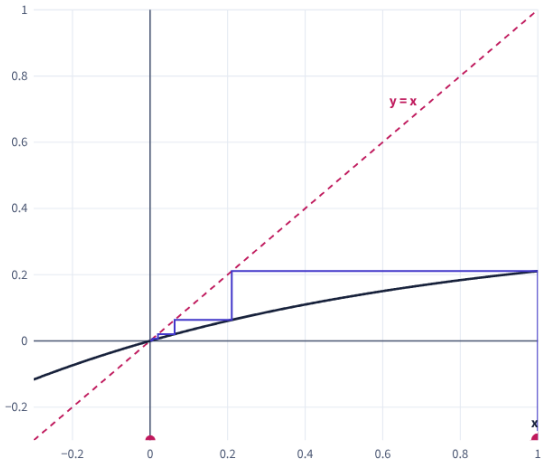
3. **The negative root.** $h(-2) \approx 0.13 > 0$ and $h(-1.5) \approx -0.34 < 0$, so $\xi_2 \in [-2, -1.5]$. There, however, $g'(x) = \frac{1}{3}e^{-x} \geq \frac{1}{3}e^{1.5} \approx 1.49 > 1$: g **is not a contraction**, and the iteration repels.
4. **Inverse rewriting.** From $3x = 1 - e^{-x}$ we get $e^{-x} = 1 - 3x$, so $x = G(x) = -\ln(1 - 3x)$. $G'(x) = \frac{3}{1-3x} > 0$, decreasing in magnitude as x decreases; the maximum on $[-2, -1.5]$ is at -1.5: $\alpha = \frac{3}{5.5} = \frac{6}{11} \approx 0.545$.
5. **Invariance.** G is increasing: $G(-2) = -\ln 7 \approx -1.946$, $G(-1.5) = -\ln 5.5 \approx -1.705$, both in $[-2, -1.5]$. ✓
6. **Iterations.** $x_0 = -1.5$, $\lvert x_1 - x_0\rvert \approx 0.2047$; $\frac{0.2047}{5/11}\alpha^n < 10^{-3}$ gives $n = 11$.

n	x_n	$\lvert x_n - x_{n-1}\rvert$	bound $\alpha^n \lvert x_1 - x_0\rvert/(1-\alpha)$	real error
0	−1.50000000	—	—	0.403814
1	−1.70474809	0.204748	0.245698	0.199066
2	−1.81062118	0.105873	0.134017	0.093193
3	−1.86126431	0.050643	0.073100	0.042549
4	−1.88461101	0.023347	0.039873	0.019203

n	x_n	$\lvert x_n - x_{n-1}\rvert$	bound $\alpha^n\lvert x_1-x_0\rvert/(1-\alpha)$	real error
5	−1.89519309	0.010582	0.021749	0.008621
6	−1.89995286	0.004760	0.011863	0.003861
7	−1.90208642	0.002134	0.006471	0.001727
8	−1.90304131	0.000955	0.003529	0.000772
9	−1.90346838	0.000427	0.001925	0.000345
10	−1.90365933	0.000191	0.001050	0.000154
11	**−1.90374469**	**0.000085**	**0.000573**	**0.000069**
12	−1.90378285	0.000038	0.000312	0.000031

7. **Result.** $\xi_1 = 0$ and $\xi_2 \approx -1.904$.

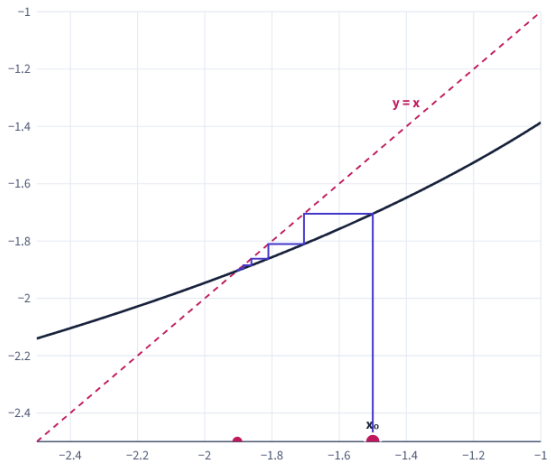
The two roots



n	x_n	$\lvert x_n - x_{n-1}\rvert$
0	1.00000000	—
1	0.21070685	0.78929315
2	0.06332951	0.14737735
3	0.02045529	0.04287422
4	0.00674917	0.01370612
5	0.00224215	0.00450702
6	0.00074655	0.00149560

After 6 steps: x_n = 0.00074655. From this x_o the sequence settles at x ≈ 0.0000000.

$g(x) = (1-e^{-x})/3$, from $x_o = 1$: attracts toward the root 0.



n	x_n	$\lvert x_n - x_{n-1}\rvert$
0	-1.50000000	—
1	-1.70474809	0.20474809
2	-1.81062118	0.10587308
3	-1.86126431	0.05064314
4	-1.88461101	0.02334670
5	-1.89519309	0.01058207
6	-1.89995286	0.00475977
7	-1.90208642	0.00213356
8	-1.90304131	0.00095489

After 8 steps: $x_n = -1.90304131$. From this x_o the sequence settles at $x \approx -1.9038137$.

$G(x) = -\ln(1-3x)$, from $x_o = -1.5$: attracts toward the negative root.

The lesson

When you rewrite inversely, that is, use the inverse function, the slope is inverted: $|G'(\xi)| = \dfrac{1}{|g'(\xi)|}$.

An iteration that repels becomes one that attracts.

Exercise 4. $x^5 + x^3 - 16 = 0$

Complete solution

1. **Isolation.** $f'(x) = 5x^4 + 3x^2 \geq 0$, zero only at 0: f is strictly increasing, so there is a single root. $f(1) = -14$, $f(2) = 24$, so $\xi \in [1,2]$.
2. **Rewriting.** See the discussion in A7. We choose $g(x) = \sqrt[5]{16 - x^3}$.
3. **α.** $|g'(x)| = \dfrac{3}{5}x^2(16 - x^3)^{-4/5}$. Both factors increase on $[1,2]$, so the maximum is at $x = 2$: $\alpha = \dfrac{3}{5} \cdot 4 \cdot 8^{-4/5} \approx 0.4547$.
4. **Invariance.** g is decreasing: $g(1) = \sqrt[5]{15} \approx 1.7188$, $g(2) = \sqrt[5]{8} \approx 1.5157$, both in $[1,2]$. ✓
5. **Iterations.** With $x_0 = 1.5$, $x_1 \approx 1.6605$, $|x_1 - x_0| \approx 0.1605$. $\dfrac{0.1605}{0.5453} \cdot 0.4547^n < 10^{-3}$: $n = 7$ gives $1.19 \cdot 10^{-3}$, not enough; $n = 8$ gives $5.4 \cdot 10^{-4}$, enough.
6. **Result.** $\xi \approx x_8 \approx 1.6337$.

n	x_n	$\lvert x_n - x_{n-1}\rvert$	bound $\alpha^n \lvert x_1 - x_0\rvert/(1-\alpha)$	real error
0	1.50000000	—	—	0.133749
1	1.66052828	0.160528	0.133865	0.026779
2	1.62758387	0.032944	0.060870	0.006165

n	x_n	$\lvert x_n - x_{n-1}\rvert$	bound $\alpha^n\lvert x_1-x_0\rvert/(1-\alpha)$	real error
3	1.63512752	0.007544	0.027679	0.001378
4	1.63343894	0.001689	0.012586	0.000310
5	1.63381888	0.000380	0.005723	0.000070
6	1.63373349	0.000085	0.002602	0.000016
7	1.63375269	0.000019	0.001183	$3.52\cdot10^{-6}$
8	**1.63374837**	**$4.31\cdot10^{-6}$**	**0.000538**	**$7.92\cdot10^{-7}$**
9	1.63374934	$9.70\cdot10^{-7}$	0.000245	$1.78\cdot10^{-7}$

S3. Newton and the tangent, solved

Level: intermediate

The column "$K\cdot(x_n-x_{n-1})^2$" is the stopping criterion from estimate ii). The bold row is the first one that satisfies it.

Problem 1. $x^3 - x - 1 = 0$ on $[1, 2]$, with $\varepsilon = 10^{-6}$.

Solution

1. $f(1) = -1, f(2) = 5$, so $f(1)f(2) < 0$. ✓
2. $f'(x) = 3x^2 - 1 \geq 2 > 0$ on $[1, 2]$; $m_1 = 2$.
3. $f''(x) = 6x > 0$, $M_2 = 12$; $K = \frac{12}{4} = 3$.
4. $f(2)f''(2) = 5 \cdot 12 > 0$, so $x_0 = 2$.

n	x_n	$f(x_n)$	$K\cdot(x_n-x_{n-1})^2$	real error
0	2.000000000000	5.000000	—	0.675282
1	1.545454545455	1.145755	0.619835	0.220737
2	1.359614915915	0.153705	0.103609	0.034897
3	1.325801345006	0.004625	0.003430	0.001083
4	1.324719049417	$4.66\cdot10^{-6}$	$3.51\cdot10^{-6}$	$1.09\cdot10^{-6}$
5	**1.324717957246**	**$4.74\cdot10^{-12}$**	**$3.58\cdot10^{-12}$**	**$1.11\cdot10^{-12}$**
6	1.324717957245	$2.22\cdot10^{-16}$	$3.71\cdot10^{-24}$	0.000000

The sequence is strictly decreasing, as the theorem predicts. $\xi \approx 1.324718$.

Problem 2. Compute $\sqrt{2}$ with $\varepsilon = 10^{-10}$.

Solution

$f(x) = x^2 - 2$ on $[1, 2]$: $f(1)f(2) = -2 < 0$; $f' = 2x$, $m_1 = 2$; $f'' = 2$, $M_2 = 2$; $K = \frac{1}{2}$. $f(2) \cdot 2 > 0$, so $x_0 = 2$.

n	x_n	$f(x_n)$	$K\cdot(x_n-x_{n-1})^2$	real error
0	2.000000000000	2.000000	—	0.585786
1	1.500000000000	0.250000	0.125000	0.085786
2	1.416666666667	0.006944	0.003472	0.002453

n	x_n	$f(x_n)$	$K\cdot(x_n-x_{n-1})^2$	real error
3	1.414215686275	$6.01\cdot10^{-6}$	$3.00\cdot10^{-6}$	$2.12\cdot10^{-6}$
4	**1.414213562375**	**$4.51\cdot10^{-12}$**	**$2.26\cdot10^{-12}$**	**$1.59\cdot10^{-12}$**
5	1.414213562373	$4.44\cdot10^{-16}$	$1.27\cdot10^{-24}$	0.000000

Problem 3. $\cos x = x$ on $[0,1]$, with $\varepsilon = 10^{-6}$.

Solution

$f(x)=\cos x - x$; $f(0)=1$, $f(1)\approx -0.46$. $f'(x)=-\sin x - 1 \le -1$, $m_1=1$. $f''(x)=-\cos x<0$ on $[0,1]$, $M_2=1$; $K=\frac{1}{2}$. We look for $f(x_0)f''(x_0)>0$: since $f''<0$, we need $f(x_0)<0$, so $x_0=1$.

n	x_n	$f(x_n)$	$K\cdot(x_n-x_{n-1})^2$	real error
0	1.000000000000	−0.459698	—	0.260915
1	0.750363867840	−0.018923	0.031159	0.011279
2	0.739112890911	−0.000046	0.000063	0.000028
3	**0.739085133385**	**$-2.85\cdot10^{-10}$**	**$3.85\cdot10^{-10}$**	**$1.70\cdot10^{-10}$**
4	0.739085133215	0.000000	$1.45\cdot10^{-20}$	0.000000

Compare: successive approximations need 16 steps to reach an error below 10^{-3} (see A6). Newton goes past 10^{-6} in 3 steps.

Problem 4. $\ln x + x - 2 = 0$ on $[1,2]$, with $\varepsilon = 10^{-6}$.

Solution

$f(1)=-1$, $f(2)=\ln 2>0$. $f'(x)=\frac{1}{x}+1\ge\frac{3}{2}=m_1$. $f''(x)=-\frac{1}{x^2}<0$, $M_2=1$; $K=\frac{1}{3}$. Since $f''<0$, we want $f(x_0)<0$: $x_0=1$. The sequence is increasing.

n	x_n	$f(x_n)$	$K\cdot(x_n-x_{n-1})^2$	real error
0	1.000000000000	−1.000000	—	0.557146
1	1.500000000000	−0.094535	0.083333	0.057146
2	1.556720935135	−0.000697	0.001072	0.000425
3	**1.557145576347**	**$-3.72\cdot10^{-8}$**	**$6.01\cdot10^{-8}$**	**$2.27\cdot10^{-8}$**
4	1.557145598998	$-2.22\cdot10^{-16}$	$1.71\cdot10^{-16}$	$2.22\cdot10^{-16}$
5	1.557145598998	0.000000	$1.64\cdot10^{-32}$	0.000000

Problem 5. The lecture example, $x^3+12x-1=0$, solved with the tangent, $\varepsilon = 10^{-4}$.

Solution

On $[0,1]$, $f''(x)=6x$ vanishes at 0: the hypothesis $f''\neq 0$ is not satisfied. We restrict: $[0.05,1]$. $f(0.05)\approx -0.3999<0$, $f(1)=12>0$. $f'(x)=3x^2+12$, $m_1=f'(0.05)=12.0075$. $f''(x)=6x>0$, $M_2=6$; $K\approx 0.25$. $x_0=1$.

n	x_n	$f(x_n)$	$K\cdot(x_n-x_{n-1})^2$	real error
0	1.000000000000	12.000000	—	0.916715
1	0.200000000000	1.408000	0.159900	0.116715

n	x_n	$f(x_n)$	$K\cdot(x_n-x_{n-1})^2$	real error
2	0.083828382838	0.006530	0.003372	0.000543
3	**0.083285197715**	**$7.40\cdot10^{-8}$**	**$7.37\cdot10^{-8}$**	**$6.16\cdot10^{-9}$**
4	0.083285191556	0.000000	$9.48\cdot10^{-18}$	0.000000
5	0.083285191556	0.000000	0.000000	0.000000

The result is the same as in A9. Newton needed 3 steps instead of 2, because the iteration g from A9 has an exceptionally small α here and a better starting point. Newton does not always win on step count.

Problem 6. Find all the roots of the equation $e^x=3x$, with $\varepsilon=10^{-6}$.

Solution

$f(x)=e^x-3x$; $f'(x)=e^x-3=0$ gives $x=\ln 3\approx 1.0986$, where $f(\ln 3)=3-3\ln 3\approx -0.296<0$. The limits are $+\infty$ at both ends: two roots. $f''(x)=e^x>0$ everywhere.

The first, on $[0,1]$: $f(0)=1$, $f(1)\approx -0.28$. $|f'|\geq 3-e\approx 0.2817=m_1$, $M_2=e$, $K\approx 4.83$. We need $f(x_0)>0$: $x_0=0$.

n	x_n	$f(x_n)$	$K\cdot(x_n-x_{n-1})^2$	real error
0	0.000000000000	1.000000	—	0.619061
1	0.500000000000	0.148721	1.206118	0.119061
2	0.610059654959	0.010362	0.058439	0.009002
3	0.618996779742	0.000074	0.000385	0.000065
4	**0.619061283355**	**$3.86\cdot10^{-9}$**	**$2.01\cdot10^{-8}$**	**$3.38\cdot10^{-9}$**
5	0.619061286736	$-2.22\cdot10^{-16}$	$5.51\cdot10^{-17}$	0.000000

The second, on $[1.2,2]$: $f(1.2)\approx -0.28$, $f(2)\approx 1.39$. $f'\geq e^{1.2}-3\approx 0.32=m_1$, $M_2=e^2$, $K\approx 11.5$. $x_0=2$.

n	x_n	$f(x_n)$	$K\cdot(x_n-x_{n-1})^2$	real error
0	2.000000000000	1.389056	—	0.487865
1	1.683518262783	0.333912	1.155973	0.171384
2	1.543481972075	0.050415	0.226324	0.031347
3	1.513488622772	0.002085	0.010382	0.001354
4	1.512137250126	$4.15\cdot10^{-6}$	0.000021	$2.70\cdot10^{-6}$
5	**1.512134551669**	**$1.65\cdot10^{-11}$**	**$8.40\cdot10^{-11}$**	**$1.07\cdot10^{-11}$**
6	1.512134551658	0.000000	$1.33\cdot10^{-21}$	0.000000

We could not choose $[1,2]$ for the second one, because $f'(\ln 3)=0$ and $\ln 3\in[1,2]$.

Problem 7. Exercise 4 from the lecture, solved with the tangent: $x^5+x^3-16=0$, $\varepsilon=10^{-3}$.

Solution

On $[1,2]$: $f'=5x^4+3x^2\geq 8=m_1$, $f''=20x^3+6x\in[26,172]$, $M_2=172$, $K=10.75$. $f(2)=24>0$, $f''>0$: $x_0=2$.

n	x_n	$f(x_n)$	$K\cdot(x_n-x_{n-1})^2$	real error
0	2.000000000000	24.000000	—	0.366251

n	x_n	$f(x_n)$	$K\cdot(x_n-x_{n-1})^2$	real error
1	1.739130434783	5.169785	0.731569	0.105381
2	1.644815352311	0.488783	0.095625	0.011066
3	1.633883705780	0.005871	0.001285	0.000135
4	**1.633749183135**	$\mathbf{8.78\cdot10^{-7}}$	$\mathbf{1.95\cdot10^{-7}}$	$\mathbf{2.01\cdot10^{-8}}$
5	1.633749163012	$2.13\cdot10^{-14}$	$4.35\cdot10^{-15}$	$4.44\cdot10^{-16}$
6	1.633749163012	$3.55\cdot10^{-15}$	$2.12\cdot10^{-30}$	0.000000

Four steps, compared with eight in S2. The bounds are, however, much more pessimistic at the start, because of the large K.

S4. Graded problems

Level: advanced

Level 1: recognition

1. Find the fixed points of $g(x)=x^2-2$.

Answer

$x^2-x-2=0$: $x=2$ and $x=-1$.

2. Is $g(x)=0.3x+5$ a contraction on $\mathbb{R}$? What is the fixed point?

Answer

Yes, $\alpha=0.3$. $p=\dfrac{5}{0.7}=\dfrac{50}{7}\approx7.143$.

3. Write Newton's formula for $f(x)=x^3-5$.

Answer

$x_n=x_{n-1}-\dfrac{x_{n-1}^3-5}{3x_{n-1}^2}=\dfrac{2x_{n-1}^3+5}{3x_{n-1}^2}$. The limit is $\sqrt[3]{5}$.

Level 2: application

4. For $g(x)=\sqrt{x+2}$ on $[0,3]$, check the hypotheses and find n for $\varepsilon=10^{-2}$, with $x_0=0$.

Solution

$g'(x)=\dfrac{1}{2\sqrt{x+2}}$ is maximal at 0: $\alpha=\dfrac{1}{2\sqrt{2}}\approx0.3536$. g is increasing, $g(0)=\sqrt{2}$, $g(3)=\sqrt{5}\approx2.236$, both in $[0,3]$. $|x_1-x_0|=\sqrt{2}$. $\dfrac{\sqrt{2}}{1-\alpha}\alpha^n\approx2.188\cdot0.3536^n<0.01$: $n=5$ gives 0.0121, $n=6$ gives 0.0043. So $n=6$. The limit is 2.

5. Is $g(x)=e^{-x}$ suitable on $[0.5,1]$? If not, correct the interval.

Solution

No: $g(1)=e^{-1}\approx 0.368\notin[0.5,1]$. Invariance fails. On $[0.3,0.8]$: g is decreasing, $g(0.8)\approx 0.449$, $g(0.3)\approx 0.741$, so invariance holds, with $\alpha=e^{-0.3}\approx 0.741$. The limit is ≈ 0.56714, called the Omega constant.

6. The equation $x^3-3x+1=0$ has three roots (P3). Find a contraction for each.

Solution

The root	Interval	g	The image	α
≈ -1.8794	$[-2,-1]$	$\sqrt[3]{3x-1}$	$\approx[-1.913,-1.587$	$4^{-2/3}\approx 0.397$
≈ 0.3473	$[0,0.5]$	$\frac{x^3+1}{3}$	$\approx[0.333,0.375]$	0.25
≈ 1.5321	$[1,2]$	$\sqrt[3]{3x-1}$	$\approx[1.260,1.710]$	$2^{-2/3}\approx 0.630$

On $[0,1]$, the function $\frac{x^3+1}{3}$ has $\sup|g'|=1$: that is why we shrank the interval.

7. Solve $x^3+x-3=0$ on $[1,2]$ with the tangent, for $\varepsilon=10^{-5}$.

Solution

$f(1)=-1, f(2)=7; f'=3x^2+1\geq 4=m_1; f''=6x\in[6,12], M_2=12, K=1.5.\ x_0=2.$

n	x_n	$f(x_n)$	$K\cdot(x_n-x_{n-1})^2$	real error
0	2.000000000000	7.000000	—	0.786588
1	1.461538461538	1.583523	0.434911	0.248127
2	1.247788154338	0.190563	0.068534	0.034376
3	1.214184564991	0.004189	0.001694	0.000773
4	**1.213412063947**	**$2.17\cdot10^{-6}$**	**$8.95\cdot10^{-7}$**	**$4.01\cdot10^{-7}$**
5	1.213411662762	$5.87\cdot10^{-13}$	$2.41\cdot10^{-13}$	$1.08\cdot10^{-13}$
6	1.213411662762	$-4.44\cdot10^{-16}$	$1.76\cdot10^{-26}$	0.000000

Level 3: argumentation

8. Let $x_0>0$, $a>0$. Show that Newton for $f(x)=\frac{1}{x}-a$ computes $\frac{1}{a}$ **without divisions**. For which x_0 does it converge?

Solution

$f'(x)=-\frac{1}{x^2}$, so $x_n=x_{n-1}+x_{n-1}^2\left(\frac{1}{x_{n-1}}-a\right)=x_{n-1}(2-a\,x_{n-1})$. Writing $e_n=1-a\,x_n$, we get $e_n=e_{n-1}^2$, so $e_n=e_0^{2^n}$. It converges exactly when $|e_0|<1$, that is $0<x_0<\frac{2}{a}$. Old processors performed division exactly this way.

9. Show that, if $K|\xi-x_0|<1$, the second estimate implies $|\xi-x_n|\leq\frac{1}{K}\left(K|\xi-x_0|\right)^{2^n}$. Apply it to $\sqrt{2}$ with $x_0=2$ and $\varepsilon=10^{-10}$.

Solution

Multiplying $e_n \le K\, e_{n-1}^2$ by K: $K\, e_n \le (K\, e_{n-1})^2$. By induction, $K\, e_n \le (K\, e_0)^{2^n}$. For $\sqrt{2}$: $K = \frac{1}{2}$, $e_0 = 2 - \sqrt{2} \approx 0.586$, $K\, e_0 \approx 0.293$. The bound $2 \cdot 0.293^{2^n}$ gives $5.8 \cdot 10^{-9}$ for $n=4$ and $1.7 \cdot 10^{-17}$ for $n=5$. So $n=5$ is guaranteed; in reality $n=4$ is already enough (the table in S3).

10. Let g be of class C^2, with $g(p) = p$ and $g'(p) = 0$. Show that the iteration has, near p, quadratic convergence.

Solution

Taylor: $x_{n+1} - p = g(x_n) - g(p) = g'(p)(x_n - p) + \frac{g''(c)}{2}(x_n - p)^2 = \frac{g''(c)}{2}(x_n - p)^2$. So $e_{n+1} \le \frac{\sup |g''|}{2} e_n^2$. Newton is exactly this case (B2).

Level 4: top performance

11. Let $f(x) = (x-\xi)^m h(x)$, $h(\xi) \neq 0$, $m \ge 2$. Show that $\varphi'(\xi) = 1 - \frac{1}{m}$. What do you conclude? How do you fix the method?

Solution

$\frac{f}{f'} = \frac{(x-\xi)h}{mh + (x-\xi)h'}$, so $\varphi(x) = x - \frac{(x-\xi)h}{mh + (x-\xi)h'}$. Dividing by $x - \xi$ and passing to the limit: $\varphi'(\xi) = 1 - \frac{1}{m}$. Since $\varphi'(\xi) \neq 0$, convergence is only linear, with factor $1 - \frac{1}{m}$ (for $m=2$, the factor is $\frac{1}{2}$).

The fix: $x_n = x_{n-1} - m\frac{f(x_{n-1})}{f'(x_{n-1})}$ has $\varphi'(\xi) = 0$ and restores quadratic convergence (L6).

12. Let $g : [a,b] \to [a,b]$ be continuous, with $|g(x) - g(y)| < |x - y|$ for $x \neq y$, but no uniform $\alpha < 1$. Show that the fixed point exists and is unique, even though g is not a contraction. Example: $\sin x$ on $[0,1]$.

Solution

Existence comes from Remark 3. Uniqueness: two fixed points would give $|p_1 - p_2| < |p_1 - p_2|$, a contradiction. For $\sin x$, $p=0$; $\frac{|\sin x|}{x} \to 1$ near 0, so no α exists. The sequence still tends to 0, but very slowly: $x_{1000} \approx 0.0546$ when $x_0 = 1$. The theorem from the lecture cannot give a bound. On $[1, \infty)$, $x + \frac{1}{x}$ shows that the compact interval is essential.

13. Let f be of class C^2, and $g_\lambda(x) = x - \lambda f(x)$. For which λ is g_λ a contraction near the simple root ξ? Which λ is optimal and what method do you get?

Solution

$g_\lambda'(\xi)=1-\lambda f'(\xi)$. The condition $|g_\lambda'(\xi)|<1$ means $0<\lambda f'(\xi)<2$. Optimal: $\lambda=\frac{1}{f'(\xi)}$, which cancels the slope. Replacing ξ with the current approximation, that is $\lambda_n=\frac{1}{f'(x_{n-1})}$, you get exactly **Newton's method**. So Newton appears as successive approximations with a parameter optimized at each step.

S5. Proofs to practice

Level: advanced

Write them on paper without looking, then compare with the indicated section. This is the list from the source, plus the necessary additions.

The result	The key idea	Where
Uniqueness of the fixed point	$\lvert p_1-p_2\rvert(1-\alpha)\le 0$	A4
$\lvert g'\rvert\le\alpha$ implies contraction	Lagrange on $[x,y]$	A4
Existence of the fixed point	$h=g-x$ changes sign	A4
Convergence and $\alpha^n(b-a)$	Iterate $\lvert x_n-p\rvert\le\alpha\lvert x_{n-1}-p\rvert$	A5
The estimate $\frac{\alpha^n}{1-\alpha}\lvert x_0-x_1\rvert$	Triangle + geometric series	A6
Newton i)	$\varphi'(\xi)=0$, so locally a contraction	B2
Newton ii)	Taylor + Lagrange	B2
The tangent theorem	Monotonic + bounded + limit	B5

S6. Self-assessment

Level: intermediate

Question 1. If $|g'(x)|\le 0.9$ on $[a,b]$, what conclusion is certain?

a) The sequence converges from every $x_0\in[a,b]$.

b) g is a 0.9-contraction on $[a,b]$.

c) g surely has a fixed point in $[a,b]$.

Question 2. The diagram shows a spiral tightening toward the fixed point. What do you know about $g'(p)$?

a) $0<g'(p)<1$

b) $g'(p)<-1$

c) $-1<g'(p)<0$

Question 3. Let $\alpha=\frac{1}{2}$ and $|x_1-x_0|=1$. How many iterations does $\frac{\alpha^n}{1-\alpha}|x_1-x_0|<10^{-3}$ require?

a) $n=11$

b) $n=10$

c) $n=20$

Question 4. f increasing and convex on $[a,b]$, with $f(a)<0<f(b)$. Where do you start the tangent method?

a) $x_0=a$

b) $x_0=b$

c) $x_0=\frac{a+b}{2}$

Question 5. For Newton, $e_{n-1}\approx 10^{-4}$ and $K\approx 1$. What do you expect e_n to be?

a) 10^{-5}

b) $5\cdot 10^{-5}$

c) 10^{-8}

Question 6. Why doesn't the tangent theorem apply directly to $x^3+12x-1$ on $[0,1]$?

a) $f''(0)=0$

b) $f(0)f(1)>0$

c) f' vanishes

Question 7. The iteration stops because $|x_{n+1}-x_n|\leq 10^{-6}$. Is it certain that $|x_{n+1}-\xi|\leq 10^{-6}$?

a) Yes, always.

b) No. If α is close to 1, the error can be much larger.

c) No, a small step means divergence.

Question 8. Newton on $(x-1)^2$. What happens?

a) It diverges.

b) It converges quadratically.

c) It converges linearly, with factor $\frac{1}{2}$.

Question 9. The iteration $x=g(x)$ repels the root, because $|g'(\xi)|>1$, and g is invertible. What do you try?

a) Another x_0, closer.

b) Iterating $x=g^{-1}(x)$.

c) Higher precision.

Question 10. Two rewritings of the same equation on the same interval, one convergent, one divergent. What is true?

a) They have the same fixed points, but different slopes at them.

b) They have different roots.

c) They are equally good.

Answers

Question	Answer	Explanation
1	b)	The criterion requires a strictly subunit α on the whole interval.
2	c)	The spiral appears when the slope is negative and its magnitude is below 1.
3	a)	$\alpha = 1/2$ and $\lvert x_1 - x_0 \rvert = 1$, so the bound is $(1/2)^n \cdot 1/(1/2) = 2 \cdot 2^{-n}$. For n = 11, it is $2^{-10} \approx 0.00098 < 10^{-3}$.
4	b)	$f(x_0)$ must have the same sign as $f''(x_0)$. Here f'' is positive, and f(b) is positive.
5	c)	If the error is of the order 10^{-4} and $K \approx 1$, the next one is of the order 10^{-8}.
6	a)	On [0, 1], $f''(x) = 6x$ vanishes at 0. The interval must be restricted.
7	b)	A small step is necessary, not sufficient. For sin x, the step is $\approx x^3/6$, much smaller than the error x.
8	c)	At a double root, $f'(\xi) = 0$, and the convergence factor becomes 1/2.
9	b)	$\lvert G'(\xi) \rvert = 1/\lvert g'(\xi) \rvert$ for the inverse function. If g repels, its inverse attracts.
10	a)	All rewritings have the same solutions: only the behavior of the iteration differs.

S7. Model test, with marking scheme

Level: advanced

Recommended time: 50 minutes. No graphing calculator. Total: 10 points.

Question 1 (2 p). Define an α-contraction. Prove that an α-contraction has at most one fixed point.

Marking scheme

- 1 p: the definition, with $\alpha \in (0,1)$ and "for all x, y".
- 1 p: the proof from A4.

Question 2 (4 p). Consider the equation $x^3 + 5x - 1 = 0$. a) Show that it has a single real root, in $[0,1]$. b) With $g(x) = \dfrac{1}{x^2+5}$, check the hypotheses of the theorem. c) Determine n for $\varepsilon = 10^{-3}$, with $x_0 = 0$, and compute x_n.

Solution and marking scheme

- **a) 1 p.** $f'=3x^2+5>0; f(0)=-1, f(1)=5.$
- **b) 1.5 p.** $u(x)=\frac{2x}{(x^2+5)^2}$; $u' \propto 5-3x^2>0$ on $[0,1]$, so $\alpha=u(1)=\frac{2}{36}=\frac{1}{18}$. Invariance: $g([0,1])=\left[\frac{1}{6},\frac{1}{5}\right]\subset[0,1].$
- **c) 1.5 p.** $x_1=\frac{1}{5}.\ \frac{1/5}{17/18}\cdot 18^{-n}<10^{-3}$: $n=1$ gives 0.0118; $n=2$ gives $6.5\cdot 10^{-4}$. So $n=2$; $x_2=\frac{1}{0.04+5}=\frac{1}{5.04}\approx 0.19841.$

Question 3 (3 p). Solve $e^x+x-2=0$ on $[0,1]$ with the tangent method, for $\varepsilon=10^{-4}$. Justify the choice of x_0.

Solution and marking scheme

- **1 p:** $f(0)=-1, f(1)=e-1>0; f'=e^x+1\geq 2=m_1; f''=e^x>0, M_2=e; K=\frac{e}{4}\approx 0.68.$
- **0.5 p:** $f(1)f''(1)>0$, so $x_0=1.$
- **1.5 p:** the iterations in the table; stop at the bold row.

n	x_n	$f(x_n)$	$K\cdot(x_n-x_{n-1})^2$	real error
0	1.000000000000	1.718282	—	0.557146
1	0.537882842740	0.250260	0.145124	0.095028
2	0.445616748527	0.007070	0.005785	0.002762
3	**0.442856724645**	**$5.94\cdot10^{-6}$**	**$5.18\cdot10^{-6}$**	**$2.32\cdot10^{-6}$**
4	0.442854401004	$4.20\cdot10^{-12}$	$3.67\cdot10^{-12}$	$1.64\cdot10^{-12}$

Question 4 (1 p). Give an example of a function and an x_0 for which Newton enters a cycle. Verify by computation.

Marking scheme

$f(x)=x^3-2x+2$, $x_0=0$: $x_1=0-\frac{2}{-2}=1$, $x_2=1-\frac{1}{1}=0.$

Lab

You write the methods and put them to the test. Below each program you find the output from actually running it.

L1. Setup

Level: beginner

The JavaScript programs in this lab use the function `print(...)`. To run them, open the HTML version of the material, where you can execute them directly, or replace `print` with `console.log` and run them in the browser console.

For Python: install Python 3 or use any online editor, save the code as `lab.py` and run `python lab.py`. For C++: `g++ -std=c++17 methods.cpp -o methods`, then run `./methods`.

The rules of a correct implementation

1. Limit the number of iterations (Remark 6).
2. Check the denominator before dividing.
3. Check that values stay finite.
4. Also return the *status* (converged or not), not just a number.
5. Do not ask for accuracy beyond the limit of the computation: in double precision, a relative $\varepsilon \approx 10^{-15}$ is the ceiling.

L2. Successive approximations

Level: beginner

Goal: write the function and reproduce the lecture example.

JavaScript program: Try it: JavaScript

```
// Method of successive approximations. Change g, x0, eps and run.
function fixedPoint(g, x0, eps, nmax) {
    let x = x0;
    for (let n = 1; n <= nmax; n++) {
        const y = g(x);
        print("n =", n, " x =", y.toFixed(10), " |step| =", Math.abs(y - x).toExponential(2));
        if (!Number.isFinite(y)) return "Divergent: non-finite value";
        if (Math.abs(y - x) <= eps) return "Converged to " + y;
        x = y;
    }
    return "Stopped: nmax reached, possible divergence";
}
print(fixedPoint(x => 1 / (x * x + 12), 0, 1e-4, 50));
```

Output:

```
n = 1  x = 0.0833333333  |step| = 8.33e-2
n = 2  x = 0.0832851359  |step| = 4.82e-5
Converged to 0.08328513591671487
```

Tasks

1. Run the code. Compare with the table in A9.
2. Change ε to 10^{-10}. How many more iterations are needed?

3. Replace with `x => (1 + Math.sin(x)) / 10` and then with `x => Math.pow(16 - x * x * x, 0.2)`, from $x_0 = 1.5$. Compare with S2.
4. Try the bad rewriting `x => 16 / (x ** 4 + x * x)`. What message do you get?
5. Add the relative criterion $\left|1 - x_{n+1}/x_n\right| \le \varepsilon$, handling the case $x_n = 0$ separately.

JavaScript program: Exercise 3: two roots

```
// Exercise 3: x = (1 - e^(-x)) / 3 has TWO roots.
// Compare the two rewritings, from two starting points.
function iter(g, x, n) {
    for (let i = 0; i < n; i++) x = g(x);
    return x;
}
const g1 = x => (1 - Math.exp(-x)) / 3;
const g2 = x => -Math.log(1 - 3 * x);
print("g1 from x0 = 1:    ", iter(g1, 1, 40));
print("g1 from x0 = -1.5: ", iter(g1, -1.5, 40));
print("g2 from x0 = -1.5: ", iter(g2, -1.5, 40));
// TASK: what do you get from g2 starting at x0 = 0.1? Explain.
```

Output:

```
g1 from x0 = 1:     0
g1 from x0 = -1.5:  0
g2 from x0 = -1.5:  -1.9038136944403785
```

Python: lab1_fixed_point.py

```
import math

def fixed_point(g, x0, eps=1e-6, nmax=100):
    """Method of successive approximations: x_(n+1) = g(x_n).
    Returns (approximation, iterations, history, status)."""
    x = x0
    history = [x0]
    for n in range(1, nmax + 1):
        try:
            x_new = g(x)
        except (ValueError, OverflowError, ZeroDivisionError):
            return x, n, history, "error: g cannot be evaluated"
        history.append(x_new)
        if not math.isfinite(x_new):
            return x_new, n, history, "divergent: non-finite value"
        if abs(x_new - x) <= eps:
            return x_new, n, history, "converged"
        x = x_new
    return x, nmax, history, "stopped: nmax reached"

# Lecture example: x^3 + 12x - 1 = 0, with g(x) = 1/(x^2 + 12)
x, n, hist, status = fixed_point(lambda x: 1 / (x**2 + 12), 0.0, 1e-4)
print(f"Lecture example: x = {x:.8f}, iterations = {n}, status = {status}")

# Bad rewriting of the same equation: x = x^3 + 13x - 1 (|g'| >= 13)
x, n, hist, status = fixed_point(lambda x: x**3 + 13*x - 1, 0.5, 1e-4, 50)
print(f"Bad rewriting: iterations = {n}, status = {status}")
```

Output:

```
Lecture example: x = 0.08328514, iterations = 2, status = converged
Bad rewriting: iterations = 7, status = error: g cannot be evaluated
```

Solution to task 5

```
const criterion = x !== 0
    ? Math.abs(1 - y / x) <= eps
    : Math.abs(y - x) <= eps;   // when x = 0, fall back to the absolute error
if (criterion) return "Converged to " + y;
```

L3. Newton's method

Level: intermediate

Goal: write the formula yourself. The code below runs, but it stops immediately, with a wrong result: the marked line does not contain the formula, as the displayed output shows.

JavaScript program: Complete the formula

```
// Complete the marked line with Newton's formula.
function newton(f, df, x0, eps, nmax) {
    let x = x0;
    for (let n = 1; n <= nmax; n++) {
        const d = df(x);
        if (d === 0) return "Failed: f'(x) = 0";
        const y = x;   // <-- to replace: y = x - ... / ...
        print("n =", n, " x =", y);
        if (Math.abs(y - x) <= eps) return "Stopped at x = " + y;
        x = y;
    }
    return "Stopped: nmax reached";
}
const f = x => x * x * x - x - 1;
const df = x => 3 * x * x - 1;
print(newton(f, df, 2, 1e-10, 30));
// Expected correct result: approximately 1.3247179572
```

Output:

```
n = 1  x = 2
Stopped at x = 2
```

Solution

```
const y = x - f(x) / d;
```

Tasks

1. With the correct formula, check the result against the table of Problem 1 in S3.
2. Run on $x^3 - 2x + 2$ from $x_0 = 0$. Confirm the cycle. The stop happens thanks to the limit n_{max}.
3. Run on $\arctan x$ from $x_0 = 1.3$ and $x_0 = 1.5$. Add a divergence check.
4. Add the variant with a numerical derivative $f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$, with $h = 10^{-6}$. Compare the number of steps.

Python: lab2_newton.py

```
def newton(f, df, x0, eps=1e-10, nmax=50):
    """Newton's method: x_n = x_(n-1) - f(x_(n-1)) / f'(x_(n-1))."""
    x = x0
    history = [x0]
    for n in range(1, nmax + 1):
        d = df(x)
        if d == 0:
            return x, n, history, "failed: f'(x) = 0"
        x_new = x - f(x) / d
        history.append(x_new)
        if abs(x_new) > 1e12:
            return x_new, n, history, "divergent"
        if abs(x_new - x) <= eps:
            return x_new, n, history, "converged"
        x = x_new
    return x, nmax, history, "stopped: nmax reached"

f  = lambda x: x**3 - x - 1
```

```
df = lambda x: 3*x**2 - 1
x, n, hist, status = newton(f, df, 2.0)
for k, v in enumerate(hist):
    print(f"x{k} = {v:.15f}")
print("status:", status, "| iterations:", n)

# A famous cycle: x^3 - 2x + 2, x0 = 0
_, n, hist, status = newton(lambda x: x**3 - 2*x + 2, lambda x: 3*x**2 - 2, 0.0, nmax=8)
print("Cycle:", hist[:6], "|", status)
```

Output:

```
x0 = 2.000000000000000
x1 = 1.545454545454545
x2 = 1.359614915915184
x3 = 1.325801345005845
x4 = 1.324719049417125
x5 = 1.324717957245858
x6 = 1.324717957244746
status: converged | iterations: 6
Cycle: [0.0, 1.0, 0.0, 1.0, 0.0, 1.0] | stopped: nmax reached
```

C++: methods.cpp

```
#include <cmath>
#include <cstdio>
#include <functional>
using Fn = std::function<double(double)>;

struct Result { double x; int iterations; bool converged; };

Result fixedPoint(Fn g, double x0, double eps, int nmax) {
    double x = x0;
    for (int n = 1; n <= nmax; ++n) {
        double y = g(x);
        if (!std::isfinite(y)) return {y, n, false};
        if (std::fabs(y - x) <= eps) return {y, n, true};
        x = y;
    }
    return {x, nmax, false};
}

Result newton(Fn f, Fn df, double x0, double eps, int nmax) {
    double x = x0;
    for (int n = 1; n <= nmax; ++n) {
        double d = df(x);
        if (d == 0.0) return {x, n, false};
        double y = x - f(x) / d;
        if (!std::isfinite(y)) return {y, n, false};
        if (std::fabs(y - x) <= eps) return {y, n, true};
        x = y;
    }
    return {x, nmax, false};
}

int main() {
    auto a = fixedPoint([](double x) { return 1.0 / (x * x + 12.0); }, 0.0, 1e-4, 100);
    std::printf("Fixed point: x = %.8f, iterations = %d, converged = %d\n", a.x, a.iterations,
a.converged);
    auto b = newton([](double x) { return x * x * x + 12.0 * x - 1.0; },
                    [](double x) { return 3.0 * x * x + 12.0; }, 1.0, 1e-10, 50);
    std::printf("Newton:      x = %.12f, iterations = %d, converged = %d\n", b.x, b.iterations,
b.converged);
}
```

Output:

```
Fixed point: x = 0.08328514, iterations = 2, converged = 1
Newton:      x = 0.083285191556, iterations = 5, converged = 1
```

L4. The tangent method, with a guaranteed stop

Level: intermediate

Goal: the program chooses x_0 by itself and uses the estimate from the lecture.

JavaScript program: Tangent method

```
// Tangent method: automatic choice of x0 and a guaranteed stop.
function tangent(f, df, d2f, a, b, m1, M2, eps) {
    if (f(a) * f(b) >= 0) return "We need f(a) * f(b) < 0";
    const x0 = f(a) * d2f(a) > 0 ? a : b;
    if (f(x0) * d2f(x0) <= 0) return "Neither endpoint is suitable";
    const K = M2 / (2 * m1);
    let x = x0;
    print("x0 =", x0, " K =", K);
    for (let n = 1; n <= 50; n++) {
        const y = x - f(x) / df(x);
        const bound = K * (y - x) * (y - x);
        print("n =", n, " x =", y.toFixed(12), " bound =", bound.toExponential(2));
        if (bound <= eps) return "xi = " + y;
        x = y;
    }
    return "Stopped at nmax";
}
print(tangent(x => Math.log(x) + x - 2, x => 1 / x + 1, x => -1 / (x * x),
              1, 2, 1.5, 1, 1e-8));
```

Output:

```
x0 = 1  K = 0.3333333333333333
n = 1  x = 1.500000000000  bound = 8.33e-2
n = 2  x = 1.556720935135  bound = 1.07e-3
n = 3  x = 1.557145576347  bound = 6.01e-8
n = 4  x = 1.557145598998  bound = 1.71e-16
xi = 1.5571455989976113
```

Tasks

1. Check the result against Problem 4 in S3.
2. Solve $x^3+x-3=0$ on $[1,2]$ and $e^x+x-2=0$ on $[0,1]$.
3. Add a monotonicity check. If the sequence is not monotonic, display a warning.
4. Write a function that estimates m_1 and M_2 on a grid of points. Careful: a grid estimate is not a proof.

Python: lab3_tangent.py

```
import math

def choose_x0(f, d2f, a, b):
    """Choose the endpoint with f(x0) * f''(x0) > 0 (tangent method)."""
    if f(a) * f(b) >= 0:
        raise ValueError("We need f(a) * f(b) < 0")
    for c in (a, b):
        if f(c) * d2f(c) > 0:
            return c
    raise ValueError("Neither endpoint satisfies f(x0) * f''(x0) > 0")

def tangent(f, df, d2f, a, b, m1, M2, eps=1e-8, nmax=50):
    """Stops using the estimate from the lecture:
       |xi - x_n| <= M2 / (2*m1) * (x_n - x_(n-1))^2 <= eps"""
    x = choose_x0(f, d2f, a, b)
```

```
    K = M2 / (2 * m1)
    steps = [x]
    for n in range(1, nmax + 1):
        x_new = x - f(x) / df(x)
        steps.append(x_new)
        if K * (x_new - x)**2 <= eps:
            return x_new, n, steps
        x = x_new
    return x, nmax, steps

# ln x + x - 2 = 0 on [1, 2]: f' = 1/x + 1 >= 3/2, |f''| = 1/x^2 <= 1
x, n, steps = tangent(lambda x: math.log(x) + x - 2,
                      lambda x: 1/x + 1,
                      lambda x: -1/x**2,
                      1.0, 2.0, m1=1.5, M2=1.0)
for k, v in enumerate(steps):
    print(f"x{k} = {v:.12f}")
print("monotone:", all(steps[i] < steps[i+1] for i in range(len(steps)-1)))
```

Output:

```
x0 = 1.000000000000
x1 = 1.500000000000
x2 = 1.556720935135
x3 = 1.557145576347
x4 = 1.557145598998
monotone: True
```

L5. Refinement: bisection followed by Newton

Level: advanced

Goal: put Remark 1 into practice. You combine the robustness of bisection with the speed of Newton. At each step you keep an interval with a sign change; if the Newton step leaves the interval, you do bisection.

JavaScript program: Safeguarded Newton

```
// Safeguarded Newton: a bisection step when Newton leaves the interval.
function hybrid(f, df, a, b, eps) {
    let fa = f(a), x = (a + b) / 2, nN = 0, nB = 0;
    if (fa * f(b) > 0) return "No sign change";
    for (let n = 1; n <= 100; n++) {
        const fx = f(x);
        if (fx === 0) return "x = " + x;
        if (fa * fx < 0) b = x; else { a = x; fa = fx; }
        const d = df(x);
        let y = d !== 0 ? x - fx / d : NaN;
        if (y > a && y < b) nN++; else { y = (a + b) / 2; nB++; }
        print("n =", n, " x =", y, " interval [", a, ",", b, "]");
        if (Math.abs(y - x) <= eps) return "x = " + y + ", Newton: " + nN + ", bisection: " +
nB;
        x = y;
    }
    return "Stopped at nmax";
}
print(hybrid(Math.atan, x => 1 / (1 + x * x), -3, 7, 1e-12));
```

Output:

```
n = 1  x = -0.5  interval [ -3 , 2 ]
n = 2  x = 0.07955951125100758  interval [ -0.5 , 2 ]
n = 3  x = -0.00033530220400547484  interval [ -0.5 , 0.07955951125100758 ]
n = 4  x = 2.5131473616567257e-11  interval [ -0.00033530220400547484 , 0.07955951125100758 ]
n = 5  x = 0  interval [ -0.00033530220400547484 , 2.5131473616567257e-11 ]
x = 0
```

On $\arctan x$, plain Newton from $x_0 = 2$ diverges, because $2 > 1.39$. The safeguarded variant reaches the root.

Tasks

1. Test on $x^3 - 2x + 2$ on $[-3, 0]$, where plain Newton can enter a cycle.
2. Count how many evaluations of f and f' you use and compare with pure bisection.
3. Explain why the interval always contains a root, whatever steps are taken.

Python: lab4_hybrid.py

```
import math

def bisection(f, a, b, steps):
    """The initial approximation from Remark 1: a few halvings."""
    if f(a) * f(b) > 0:
        raise ValueError("f(a) and f(b) must have opposite signs")
    for _ in range(steps):
        c = (a + b) / 2
        if f(a) * f(c) <= 0:
            b = c
        else:
            a = c
    return a, b

def hybrid(f, df, a, b, eps=1e-12, nmax=100):
    """Safeguarded Newton: if the step leaves the interval or f' = 0,
    take a bisection step. The interval always brackets the root."""
    fa = f(a)
    if fa * f(b) > 0:
        raise ValueError("f(a) and f(b) must have opposite signs")
    x = (a + b) / 2
    newton_steps, bisection_steps = 0, 0
    for n in range(1, nmax + 1):
        fx = f(x)
        if fx == 0:
            return x, n, newton_steps, bisection_steps
        if fa * fx < 0:
            b = x
        else:
            a, fa = x, fx
        d = df(x)
        y = x - fx / d if d != 0 else a - 1
        if a < y < b:
            newton_steps += 1
        else:
            y = (a + b) / 2
            bisection_steps += 1
        if abs(y - x) <= eps:
            return y, n, newton_steps, bisection_steps
        x = y
    return x, nmax, newton_steps, bisection_steps

print("Bisection, 4 steps on [0, 1] for x^3 + 12x - 1:",
      bisection(lambda x: x**3 + 12*x - 1, 0, 1, 4))
# arctan x: plain Newton diverges from x0 = 2 (midpoint of [-3, 7]), the hybrid does not
print("Hybrid, arctan on [-3, 7]: x = %.2e, iterations = %d, Newton = %d, bisection = %d"
      % hybrid(math.atan, lambda x: 1/(1 + x*x), -3, 7))
```

Output:

```
Bisection, 4 steps on [0, 1] for x^3 + 12x - 1: (0.0625, 0.125)
Hybrid, arctan on [-3, 7]: x = 0.00e+00, iterations = 6, Newton = 4, bisection = 1
```

L6. Measuring the order of convergence

Level: top performance

If $e_{n+1} \approx C\, e_n^p$, then the **order** p can be estimated from three consecutive errors:

$$p \approx \frac{\ln\left(e_{n+1}/e_n\right)}{\ln\left(e_n/e_{n-1}\right)}.$$

JavaScript program: Estimate the order

```
// Estimated order of convergence from a sequence of iterates.
function order(seq, r) {
    const e = seq.map(x => Math.abs(x - r));
    const p = [];
    for (let i = 1; i + 1 < e.length; i++)
        if (e[i - 1] > 0 && e[i] > 0 && e[i + 1] > 1e-14)
            p.push(+(Math.log(e[i + 1] / e[i]) / Math.log(e[i] / e[i - 1])).toFixed(3));
    return p;
}
function newtonSeq(f, df, x, n, m = 1) {
    const v = [x];
    for (let i = 0; i < n; i++) { const d = df(x); if (d === 0) break; x -= m * f(x) / d;
v.push(x); }
    return v;
}
const f = x => (x - 1) ** 2 * (x + 2), df = x => 2 * (x - 1) * (x + 2) + (x - 1) ** 2;
print("Double root, Newton:    ", order(newtonSeq(f, df, 2, 14), 1));
print("Double root, m = 2:     ", order(newtonSeq(f, df, 2, 5, 2), 1));
const g = x => Math.cos(x) - x, dg = x => -Math.sin(x) - 1;
print("Simple root, Newton:    ", order(newtonSeq(g, dg, 0.2, 5), 0.7390851332151607));
```

Output:

```
Double root, Newton:     [1.06,1.044,1.029,1.017,1.009,1.005,1.002,1.001,1.001,1,1,1,1]
Double root, m = 2:      [1.84,1.987,2]
Simple root, Newton:     [2.428,1.969,2]
```

Expected values: $p \approx 1$ for fixed point and multiple roots, $p \approx 2$ for Newton with a simple root. The first values may be far from the limit, and the last are affected by the finite precision of the computer.

Tasks

1. For Newton on the double root, explain why $e_{n+1}/e_n \to \frac{1}{2}$.
2. See $f(x) = (x-1)^3$, with $m = 1$ and with $m = 3$. What is the factor? Compare with problem 11 in S4.
3. Find a function for which the Newton order is 3, that is $f''(\xi) = 0$. Hint: try $\sin x$ around 0.

Python: lab5_order.py

```
import math

def estimated_order(xs, root):
    """p_n = ln(e_(n+1)/e_n) / ln(e_n/e_(n-1))"""
    e = [abs(x - root) for x in xs]
    result = []
    for i in range(1, len(e) - 1):
        if e[i-1] > 0 and e[i] > 0 and e[i+1] > 1e-14:
            result.append(math.log(e[i+1]/e[i]) / math.log(e[i]/e[i-1]))
    return result

def fp_seq(g, x, n):
    v = [x]
```

```
    for _ in range(n):
        x = g(x); v.append(x)
    return v

def newton_seq(f, df, x, n, m=1):
    v = [x]
    for _ in range(n):
        d = df(x)
        if d == 0:
            break
        x = x - m * f(x) / d; v.append(x)
    return v

r = 0.7390851332151607
print("Fixed point, cos x:",
      [round(p, 3) for p in estimated_order(fp_seq(math.cos, 1.0, 12), r)][-3:])
print("Newton, cos x - x:",
      [round(p, 3) for p in estimated_order(newton_seq(lambda x: math.cos(x) - x, lambda x:
-math.sin(x) - 1, 0.2, 5), r)])
f, df = (lambda x: (x-1)**2 * (x+2)), (lambda x: 2*(x-1)*(x+2) + (x-1)**2)
print("Newton, double root:",
      [round(p, 3) for p in estimated_order(newton_seq(f, df, 2.0, 12), 1.0)][-3:])
print("Modified Newton (m = 2), double root:",
      newton_seq(f, df, 2.0, 5, m=2)[:5])
```

Output:

```
Fixed point, cos x: [1.027, 0.982, 1.012]
Newton, cos x - x: [2.428, 1.969, 2.0]
Newton, double root: [1.001, 1.0, 1.0]
Modified Newton (m = 2), double root: [2.0, 1.1111111111111112, 1.0019493177387915,
1.0000006326899509, 1.0000000000000666]
```

L7. Real applications

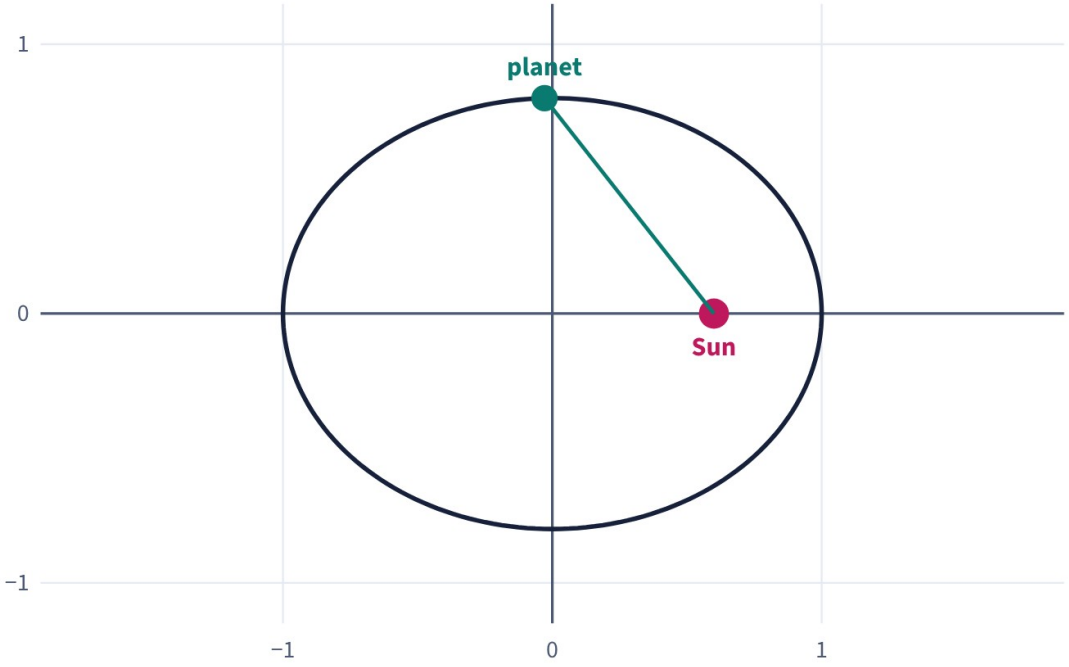
Level: advanced

Kepler's equation: where is the planet?

The position on an elliptical orbit is obtained from **Kepler's equation**, $E - e\sin E = M$, where e is the eccentricity, M grows uniformly in time, and the unknown is E. There is no elementary formula for E.

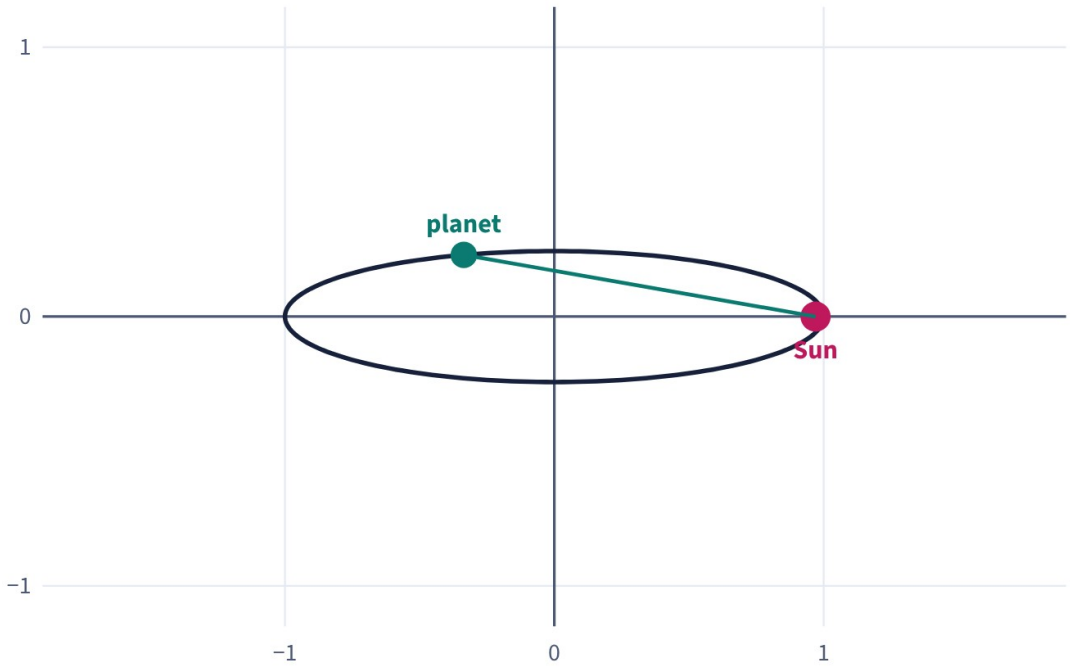
The equation already has the form $E = g(E)$, with $g(E) = M + e\sin E$ and $|g'(E)| = e|\cos E| \le e < 1$. The theorem applies on the whole axis, and α **is exactly the eccentricity**.

Orbit simulator



Successive approximations, E = M + e·sin E (α = e)	9	1.5997485482
Newton	6	1.5997485482

Orbit with e = 0.6 and M = 1.



Successive approximations, E = M + e·sin E (α = e)	25	1.9135709611
Newton	6	1.9135709611

Orbit with e = 0.97 and M = 1.

Python: lab6_kepler.py

```
import math

def kepler_fp(M, e, eps=1e-12, nmax=100000):
    E = M
    for n in range(1, nmax + 1):
        new = M + e * math.sin(E)
        if abs(new - E) <= eps:
            return new, n
        E = new
    return E, nmax

def kepler_newton(M, e, eps=1e-12, nmax=50):
    E = M if e < 0.8 else math.pi
    for n in range(1, nmax + 1):
        new = E - (E - e*math.sin(E) - M) / (1 - e*math.cos(E))
        if abs(new - E) <= eps:
            return new, n
        E = new
    return E, nmax

print(f"{'e':>5} {'E (fixed point)':>16} {'iter':>6} {'E (Newton)':>16} {'iter':>5}")
for e in [0.0167, 0.2056, 0.5, 0.9, 0.967]:
    E1, n1 = kepler_fp(1.0, e)
    E2, n2 = kepler_newton(1.0, e)
    print(f"{e:5.4f} {E1:16.12f} {n1:6d} {E2:16.12f} {n2:5d}")

def irr(flows, r=0.1, eps=1e-10):
    """Internal rate of return: NPV(r) = sum CF_t / (1+r)^t = 0"""
    for _ in range(100):
        v = sum(c / (1+r)**t for t, c in enumerate(flows))
```

```
        d = sum(-t * c / (1+r)**(t+1) for t, c in enumerate(flows))
        new = r - v / d
        if abs(new - r) <= eps:
            return new
        r = new
    return r

print("IRR for [-1000, 300, 400, 500]: %.4f%%" % (100*irr([-1000, 300, 400, 500])))
```

Output:

```
    e  E (fixed point)   iter       E (Newton)  iter
0.0167   1.014179087165      6   1.014179087165     3
0.2056   1.190944767223     12   1.190944767223     4
0.5000   1.498701133518     10   1.498701133518     5
0.9000   1.862086686874     21   1.862086686875     6
0.9670   1.911436976490     25   1.911436976490     6
IRR for [-1000, 300, 400, 500]: 8.8963%
```

Eccentricities in the table: 0.0167 is Earth's orbit, 0.2056 Mercury's, 0.967 Halley's comet (approximate, rounded values). The number of steps does not increase strictly with e, because it also depends on the real slope $e\cos E$ at the solution, not just on the bound.

Internal rate of return

An investment of 1000 returns 300, 400 and 500 over the next three years. What annual interest rate r is equivalent? We have to solve the equation:

$$-1000+\frac{300}{1+r}+\frac{400}{(1+r)^2}+\frac{500}{(1+r)^3}=0.$$

The program above gives $r\approx 8.90\,\%$. This is the equation that the IRR function in a spreadsheet solves.

Tasks

1. Solve Kepler's equation for $e=0.5$ and $M=2$, with both methods.
2. Compute how many iterations the a priori bound requires for $e=0.967$, $\varepsilon=10^{-12}$, and compare with the table.
3. Compute the internal rate for a loan: you receive 10 000 and pay 3 000 per year for 4 years.

L8. Accelerating convergence

Level: top performance

If the sequence converges linearly, $x_{n+1}-p\approx\lambda\left(x_n-p\right)$, two consecutive relations let you eliminate λ and estimate the limit directly. You get **Aitken's Δ² formula**:

$$\hat{x}_n=x_n-\frac{\left(x_{n+1}-x_n\right)^2}{x_{n+2}-2\,x_{n+1}+x_n}.$$

If you restart the iteration from $\hat{x}_n$ you get **Steffensen's** method: quadratic convergence, **without derivatives**, using only g.

Python: lab7_aitken.py

```
import math

def steffensen(g, x, eps=1e-12, nmax=50):
    """Aitken acceleration applied repeatedly; uses no derivative."""
    for n in range(1, nmax + 1):
```

```
        x1 = g(x); x2 = g(x1)
        den = x2 - 2*x1 + x
        if den == 0:
            return x2, n
        new = x - (x1 - x)**2 / den
        if abs(new - x) <= eps:
            return new, n
        x = new
    return x, nmax

x = 1.0; n = 0
while abs(math.cos(x) - x) > 1e-12:
    x = math.cos(x); n += 1
print(f"Plain fixed point, g = cos: {x:.12f} in {n} iterations")
print("Steffensen, g = cos: %.12f in %d iterations" % steffensen(math.cos, 1.0))
```

Output:

```
Plain fixed point, g = cos: 0.739085133216 in 68 iterations
Steffensen, g = cos: 0.739085133215 in 5 iterations
```

From 68 iterations to 5. Each Steffensen iteration, however, uses two evaluations of g, so the fair comparison is 68 versus 10 evaluations.

Tasks

1. Test Steffensen on $g(x)=\frac{16}{x^4+x^2}$, the divergent rewriting from A7. What do you observe?
2. Compare the number of evaluations with Newton, on the same equation.

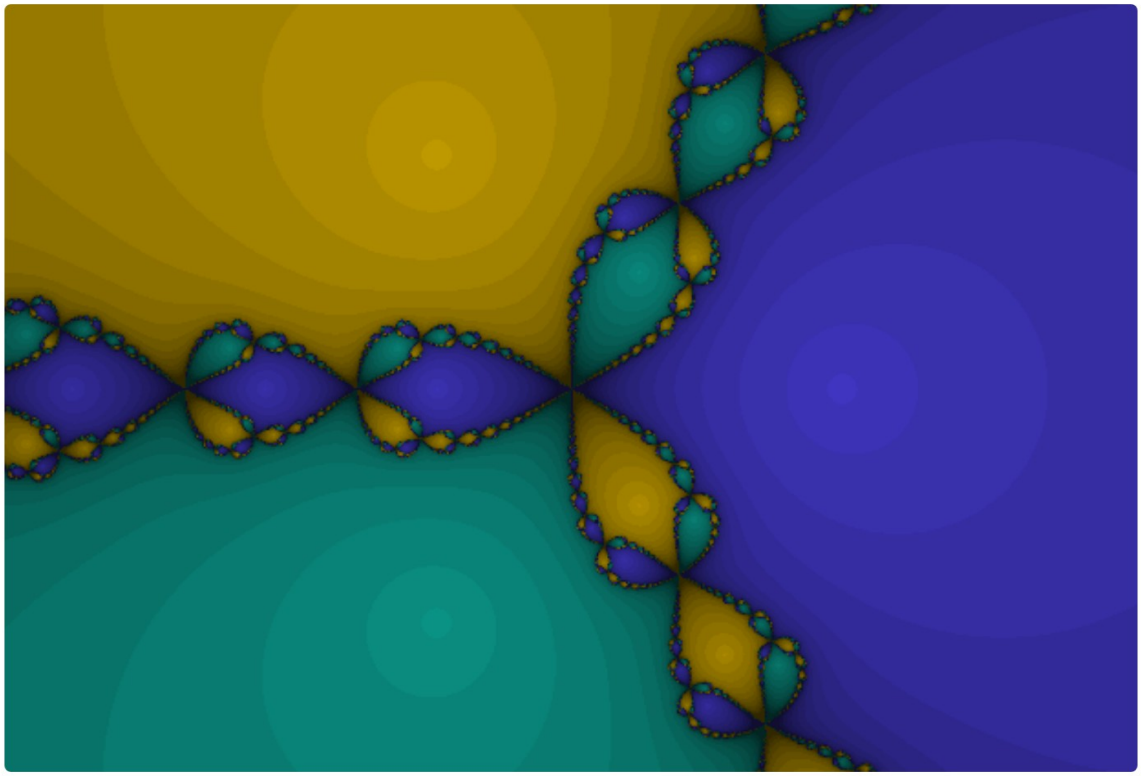
What you should observe in task 1

Near the root, Steffensen can converge even when the simple iteration diverges, because it does not depend on the condition $|g'(p)|<1$. However, convergence is no longer globally guaranteed, and it may fail from distant starting points.

L9. Beyond the real line and a mini-project

Level: top performance

Newton's formula also works for complex numbers. For $z^3-1=0$ there are three roots. We color each starting point by the root it reaches. The boundary between the basins is a **fractal**, the same instability you saw on the real line in B4. This content goes beyond the source, which is limited to $\mathbb{R}$; it is offered as an extension.



Newton basins for z^3-1. The color shows the root reached, the shade shows the number of steps.

Mini-project: an equation-solving library

Create a module that includes:

1. the functions `fixed_point`, `newton`, `tangent`, `hybrid`, all with the same result format (approximation, iterations, status, history);
2. automated tests for the four exercises from the lecture, comparing with the solutions in S2;
3. tests for failure situations: cycle, divergence, $f'=0$;
4. a chart of error versus iteration, on a logarithmic scale;
5. a short report: which method you would recommend for each equation and why.

Criterion	Points
The methods are correct and return the status	30
The hypotheses are checked and reported	20
The tests also cover failure cases	20
Error charts and the estimated order	15
The report is well argued	15

At the end

Review and coverage check.

Z1. Review sheet

Level: beginner

Successive approximations	*Newton / tangent*
$x_{n+1}=g(x_n)$ **Hypotheses:** $g:[a,b]\to[a,b]$, $\lvert g'\rvert\le\alpha<1$. **Errors:** $\lvert x_n-p\rvert\le\alpha^n(b-a)$ $\lvert x_n-p\rvert\le\dfrac{\alpha^n}{1-\alpha}\lvert x_0-x_1\rvert$ **Geometry:** the intersection with $y=x$. **Speed:** linear.	$x_n=x_{n-1}-\dfrac{f(x_{n-1})}{f'(x_{n-1})}$ **Hypotheses:** $f(a)f(b)<0$; f', $f''\neq0$; $f(x_0)f''(x_0)>0$. **Errors:** $\lvert\xi-x_n\rvert\le\dfrac{M_2}{2m_1}(x_n-x_{n-1})^2$ $\lvert\xi-x_n\rvert\le\dfrac{M_2}{2m_1}(\xi-x_{n-1})^2$ **Geometry:** the tangent crosses Ox at x_n. **Speed:** quadratic, if $f'(\xi)\neq0$.

The most common mistakes

- Checking $\lvert g'\rvert<1$ but forgetting $g([a,b])\subseteq[a,b]$.
- Finding one solution and assuming it is the only one (exercise 3).
- Checking $\alpha<1$ at a single point, not on the whole interval.
- Computing $\sup g'$ instead of $\sup\lvert g'\rvert$.
- Forgetting that the logarithm of α is negative and flips the inequality.
- Starting the tangent method from the wrong endpoint.
- Not noticing that f'' vanishes on the interval.
- Believing that a small step means a small error.
- Working in degrees instead of radians.

Z2. Coverage matrix

Level: beginner

Each element of the source document, with the section that covers it.

Element in the source	Theory	Practice
Definition 1: fixed point	A2	S4, S6
Definition 2: α-contraction	A3	S4, S7
Remarks 1, 2, 3, with proofs	A4	S5, S7
The theorem of successive approximations	A5	S2, L2

Element in the source	Theory	Practice
Remarks: refinement with secant or bisection	A8	L5
Remarks: $f(x)=0$ becomes $x=g(x)$; Banach	A7	S2
Remarks: the geometric interpretation	A2, A5	the figures in the lecture
Remarks: the influence of α	A5	L7
Remarks: stopping criteria, relative error	A8	L2
Remarks: limiting the iterations	A8	L1–L3
Remarks: $g:M\to M$, M closed	A6, A8	S2, ex. 2
The example $x^3+12x-1=0$; Rolle's sequence	P3, A9	L2
The exercises: the four equations		S2
Newton: figure and formula	B1	S3, L3
Newton's theorem, i)	B2	S5
Newton's theorem, ii): the estimates	B2, B3	S3, L4
The disadvantage: choosing x_0	B4	L3
The tangent theorem	B5	S3, L4

What I added or clarified relative to the source

- **Proofs missing from the source:** the theorem of successive approximations, the estimate in Remark 7, Newton i) and ii), the tangent theorem.
- **Clarification:** for Remark 2 in the source, the theorem also requires $g([a,b])\subseteq[a,b]$ (A7). This check is missing from the lecture example, so I added it (A9).
- **Clarification:** the word "roots" in the example: the equation has a single real root.
- **Note on the exercises:** exercise 3 has two solutions, and the given form is not a contraction near the negative solution (S2).
- **The estimates in Newton ii)** assume that the terms of the sequence stay in $[a,b]$, where m_1 and M_2 are defined (B2).
- **Extensions, clearly marked:** order of convergence, multiple roots, Aitken and Steffensen, applications, the fractal. They go beyond the source and are not needed to cover it.

Z3. Glossary

Level: beginner

Term	In short
Fixed point	p with $g(p)=p$.
Contraction	A function that shrinks all distances by at least the factor $\alpha<1$.

Term	In short
First bisector	The line $y = x$.
Absolute, relative error	$\left\|x_n - \xi\right\|$, respectively scaled by the size of the value.
A priori, a posteriori	Computed before iterating, respectively from the iterations already done.
Linear convergence	$e_n \leq \alpha\, e_{n-1}$, with $\alpha < 1$.
Quadratic convergence	$e_n \leq K\, e_{n-1}^2$.
Simple, multiple root	$f'(\xi) \neq 0$, respectively $f'(\xi) = 0$.
Basin of attraction	The set of points x_0 from which the sequence reaches a given root.
Class C^2	Twice differentiable, with f'' continuous.
Root isolation	Finding one interval for each root.